STATE OF NEW YORK
DEPARTMENT OF PUBLIC SERVICE

Case 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

Case 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.

Case 18-M-0270 - Petition for Approval Pursuant to Public Service Law Section 113(2), of a Proposed Allocation of Certain Tax Refunds between KeySpan Gas East Corporation d/b/a National Grid and Ratepayers.

STAFF REPLY STATEMENT

APPENDIX 1

Appendix 1 to the Staff Reply Statement contains information regarding bill impacts for The Brooklyn Union Gas Company d/b/a National Grid NY's service classification 21 Rate 3.

Calculation of SC 21 Rate 3 savings from receiving average increase

Line	Description	SC-21 DG Large	SC-21 DG Large	Difference
1	**Step 1- Historic Rate of Return**			
2	**Delivery revenue present rates**	$ 2,215,922	$ 2,215,922	
3	Tax Credit	$ (45,885)	$ (45,885)	
4	ETIP	$ -	$ -	
5	MFC at Present rates	$ 174,771	$ 174,771	
6	**Delivery revenue, pro forma**	$ 2,344,808	$ 2,344,808	
7				
8	Return at Historic Rates	(25.9%)	(25.9%)	
9	Relative Return at Present Rates	(5.28)	(5.28)	
10				
11	**Step 2- Tolerance Band for Return at Present Rates**			
12	Top of Band			
13	Bottom of Band			
14	Within Band			
15				
16	**Step 3- Position of Classes / Tolerance Band**	DG	DG	
17		Average	Under	
18	**Step 4- Increase for Target Revenue**			
19	**Proposed Increase %**	**6.60%**	**8.70%**	
20	Change in Pro Forma	**154,788**	**203,998**	**$ 49,210**
				24%

Calculation of SC 21 Rate 3 revenue deficiency

Line	Description	SC-21 DG Large	
1	Net (Loss) Historic	$	(1,826,000)
2	Additional revenue	$	1,826,000
3	Net income with Increase	$	-
4	Rate Base	$	8,811,000
5	Rate of return		0.00%
6			
7	Net (Loss) Historic	$	(1,826,000)
8	Additional revenue	**$**	**2,150,000**
9	Net income with Increase	$	324,000
10	Rate Base	$	8,811,000
11	Rate of return		3.68%
12	System average return		4.92%
13	Relative return		0.75

Staff calculations of Rate Year 2 bill impacts to BSC Owner SC 21-3

		Total Bill					Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-21	140,016	$ 715,971	$ 724,729	$ 8,759	1.22%	$ 77,136	$ 77,136	$ (0)	0.00%
May-21	129,717	$ 666,724	$ 674,838	$ 8,114	1.22%	$ 74,880	$ 74,879	$ (0)	0.00%
Jun-21	155,729	$ 791,146	$ 800,887	$ 9,742	1.23%	$ 80,620	$ 80,620	$ (0)	0.00%
Jul-21	116,977	$ 605,772	$ 613,090	$ 7,317	1.21%	$ 72,055	$ 72,055	$ (0)	0.00%
Aug-21	165,015	$ 835,599	$ 854,063	$ 18,464	2.21%	$ 82,705	$ 94,166	$ 11,461	13.86%
Sep-21	111,761	$ 580,791	$ 594,479	$ 13,688	2.36%	$ 70,873	$ 79,817	$ 8,945	12.62%
Oct-21	67,335	$ 368,271	$ 377,975	$ 9,703	2.63%	$ 61,050	$ 67,896	$ 6,846	11.21%
Nov-21	74,817	$ 410,952	$ 421,326	$ 10,374	2.52%	$ 69,593	$ 76,793	$ 7,199	10.34%
Dec-21	82,299	$ 447,465	$ 458,511	$ 11,045	2.47%	$ 71,970	$ 79,522	$ 7,553	10.49%
Jan-22	82,299	$ 447,465	$ 458,511	$ 11,045	2.47%	$ 71,970	$ 79,522	$ 7,553	10.49%
Feb-22	82,299	$ 447,386	$ 458,432	$ 11,045	2.47%	$ 71,890	$ 79,443	$ 7,553	10.51%
Mar-22	67,335	$ 374,491	$ 384,195	$ 9,703	2.59%	$ 67,270	$ 74,116	$ 6,846	10.18%
Annual Total	1,275,599	$ 6,692,033	$ 6,821,035	$ 129,002	1.93%	$ 872,012	$ 935,966	$ 63,954	7.33%

Staff calculations of Rate Year 3 bill impacts to BSC Owner SC 21-3

		Total Bill					Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-22	140,016	$ 724,729	$ 747,663	$ 22,934	3.16%	$ 77,136	$ 102,887	$ 25,750	33.38%
May-22	129,717	$ 674,838	$ 696,473	$ 21,635	3.21%	$ 74,879	$ 99,124	$ 24,244	32.38%
Jun-22	155,729	$ 800,887	$ 825,803	$ 24,916	3.11%	$ 80,620	$ 108,668	$ 28,048	34.79%
Jul-22	116,977	$ 613,090	$ 633,118	$ 20,028	3.27%	$ 72,055	$ 94,437	$ 22,381	31.06%
Aug-22	165,015	$ 857,383	$ 872,008	$ 14,625	1.71%	$ 94,166	$ 112,111	$ 17,945	19.06%
Sep-22	111,761	$ 596,727	$ 607,153	$ 10,425	1.75%	$ 79,817	$ 92,491	$ 12,674	15.88%
Oct-22	67,335	$ 379,329	$ 386,251	$ 6,922	1.82%	$ 67,896	$ 76,172	$ 8,276	12.19%
Nov-22	74,817	$ 422,831	$ 430,420	$ 7,589	1.79%	$ 76,793	$ 85,886	$ 9,094	11.84%
Dec-22	82,299	$ 460,166	$ 468,353	$ 8,186	1.78%	$ 79,522	$ 89,364	$ 9,842	12.38%
Jan-23	82,299	$ 460,166	$ 468,353	$ 8,186	1.78%	$ 79,522	$ 89,364	$ 9,842	12.38%
Feb-23	82,299	$ 460,087	$ 468,273	$ 8,186	1.78%	$ 79,443	$ 89,285	$ 9,842	12.39%
Mar-23	67,335	$ 385,549	$ 392,540	$ 6,991	1.81%	$ 74,116	$ 82,461	$ 8,345	11.26%
Annual Total	1,275,599	$ 6,835,783	$ 6,996,407	$ 160,623	2.35%	$ 935,966	$ 1,122,250	$ 186,284	19.90%

*Rate Year 3 Bill Impacts are affected by Compression of Rate Year 2 Rates.

Staff calculations of Rate Year 4 bill impacts to BSC Owner SC 21-3

		Total Bill					Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-23	140,016	$ 747,663	$ 777,160	$ 29,497	3.95%	$ 102,887	$ 132,384	$ 29,497	28.67%
May-23	129,717	$ 696,473	$ 723,800	$ 27,327	3.92%	$ 99,124	$ 126,451	$ 27,327	27.57%
Jun-23	155,729	$ 825,803	$ 858,610	$ 32,807	3.97%	$ 108,668	$ 141,475	$ 32,807	30.19%
Jul-23	116,977	$ 633,118	$ 657,761	$ 24,643	3.89%	$ 94,437	$ 119,080	$ 24,643	26.10%
Aug-23	165,015	$ 872,008	$ 906,771	$ 34,763	3.99%	$ 112,111	$ 146,874	$ 34,763	31.01%
Sep-23	111,761	$ 607,153	$ 630,697	$ 23,544	3.88%	$ 92,491	$ 116,035	$ 23,544	25.46%
Oct-23	67,335	$ 386,251	$ 400,436	$ 14,185	3.67%	$ 76,172	$ 90,357	$ 14,185	18.62%
Nov-23	74,817	$ 430,420	$ 446,949	$ 16,529	3.84%	$ 85,886	$ 102,416	$ 16,529	19.25%
Dec-23	82,299	$ 468,353	$ 486,535	$ 18,182	3.88%	$ 89,364	$ 107,547	$ 18,182	20.35%
Jan-24	82,299	$ 468,353	$ 486,535	$ 18,182	3.88%	$ 89,364	$ 107,547	$ 18,182	20.35%
Feb-24	82,299	$ 468,273	$ 486,456	$ 18,182	3.88%	$ 89,285	$ 107,467	$ 18,182	20.36%
Mar-24	67,335	$ 392,540	$ 407,416	$ 14,876	3.79%	$ 82,461	$ 97,337	$ 14,876	18.04%
Annual Total	1,275,599	$ 6,996,407	$ 7,269,126	$ 272,720	3.90%	$ 1,122,250	$ 1,394,970	$ 272,720	24.30%

Staff calculations of Rate Year 2 bill impacts to BSC Owner SC 3

		Total Bill				Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-21	52,200	$ 41,888	$ 44,104	$ 2,217	5.29%	$ 18,071	$ 19,864	$ 1,794	9.93%
May-21	41,968	$ 33,709	$ 35,487	$ 1,778	5.28%	$ 14,561	$ 15,999	$ 1,438	9.88%
Jun-21	29,404	$ 23,667	$ 24,907	$ 1,241	5.24%	$ 10,251	$ 11,253	$ 1,002	9.78%
Jul-21	16,965	$ 13,724	$ 14,432	$ 708	5.16%	$ 5,984	$ 6,554	$ 570	9.53%
Aug-21	8,549	$ 6,997	$ 7,345	$ 348	4.97%	$ 3,096	$ 3,375	$ 278	8.99%
Sep-21	5,206	$ 4,325	$ 4,529	$ 204	4.73%	$ 1,950	$ 2,112	$ 162	8.32%
Oct-21	4,932	$ 4,106	$ 4,299	$ 193	4.69%	$ 1,856	$ 2,008	$ 153	8.23%
Nov-21	4,820	$ 4,016	$ 4,204	$ 188	4.68%	$ 1,817	$ 1,966	$ 149	8.19%
Dec-21	4,904	$ 4,083	$ 4,275	$ 192	4.69%	$ 1,846	$ 1,998	$ 152	8.22%
Jan-22	12,958	$ 10,521	$ 11,057	$ 536	5.10%	$ 4,609	$ 5,040	$ 431	9.36%
Feb-22	24,434	$ 19,694	$ 20,722	$ 1,028	5.22%	$ 8,546	$ 9,375	$ 830	9.71%
Mar-22	44,791	$ 35,965	$ 37,865	$ 1,899	5.28%	$ 15,529	$ 17,066	$ 1,536	9.89%
Annual Total	251,131	$ 202,695	$ 213,226	$ 10,531	5.20%	$ 88,115	$ 96,611	$ 8,496	9.64%

Staff calculations of Rate Year 3 bill impacts to BSC Owner SC 3

		Total Bill				Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-22	52,200	$ 44,104	$ 45,287	$ 1,183	2.68%	$ 19,864	$ 21,047	$ 1,183	5.95%
May-22	41,968	$ 35,487	$ 36,439	$ 951	2.68%	$ 15,999	$ 16,951	$ 951	5.95%
Jun-22	29,404	$ 24,907	$ 25,574	$ 667	2.68%	$ 11,253	$ 11,920	$ 667	5.93%
Jul-22	16,965	$ 14,432	$ 14,818	$ 386	2.67%	$ 6,554	$ 6,940	$ 386	5.89%
Aug-22	8,549	$ 7,345	$ 7,540	$ 195	2.66%	$ 3,375	$ 3,570	$ 195	5.79%
Sep-22	5,206	$ 4,529	$ 4,649	$ 120	2.64%	$ 2,112	$ 2,232	$ 120	5.67%
Oct-22	4,932	$ 4,299	$ 4,412	$ 113	2.64%	$ 2,008	$ 2,122	$ 113	5.65%
Nov-22	4,820	$ 4,204	$ 4,315	$ 111	2.64%	$ 1,966	$ 2,077	$ 111	5.64%
Dec-22	4,904	$ 4,275	$ 4,388	$ 113	2.64%	$ 1,998	$ 2,111	$ 113	5.65%
Jan-23	12,958	$ 11,057	$ 11,353	$ 295	2.67%	$ 5,040	$ 5,335	$ 295	5.85%
Feb-23	24,434	$ 20,722	$ 21,276	$ 555	2.68%	$ 9,375	$ 9,930	$ 555	5.92%
Mar-23	44,791	$ 37,865	$ 38,880	$ 1,015	2.68%	$ 17,066	$ 18,081	$ 1,015	5.95%
Annual Total	251,131	$ 213,226	$ 218,931	$ 5,705	2.68%	$ 96,611	$ 102,315	$ 5,705	5.90%

*Rate Year 3 Bill Impacts are affected by Compression of Rate Year 2 Rates.

Staff calculations of Rate Year 4 bill impacts to BSC Owner SC 3

		Total Bill				Delivery Bill			
Month	Usage (Dth)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)	Current Bill	Bill @ JP rates	Impact ($)	Impacts (%)
Apr-23	52,200	$ 45,287	$ 48,220	$ 2,933	6.48%	$ 21,047	$ 23,980	$ 2,933	13.94%
May-23	41,968	$ 36,439	$ 38,798	$ 2,359	6.47%	$ 16,951	$ 19,310	$ 2,359	13.92%
Jun-23	29,404	$ 25,574	$ 27,228	$ 1,654	6.47%	$ 11,920	$ 13,574	$ 1,654	13.88%
Jul-23	16,965	$ 14,818	$ 15,774	$ 956	6.45%	$ 6,940	$ 7,896	$ 956	13.78%
Aug-23	8,549	$ 7,540	$ 8,024	$ 484	6.42%	$ 3,570	$ 4,054	$ 484	13.57%
Sep-23	5,206	$ 4,649	$ 4,946	$ 297	6.38%	$ 2,232	$ 2,528	$ 297	13.30%
Oct-23	4,932	$ 4,412	$ 4,694	$ 281	6.38%	$ 2,122	$ 2,403	$ 281	13.26%
Nov-23	4,820	$ 4,315	$ 4,590	$ 275	6.38%	$ 2,077	$ 2,352	$ 275	13.25%
Dec-23	4,904	$ 4,388	$ 4,668	$ 280	6.38%	$ 2,111	$ 2,390	$ 280	13.26%
Jan-24	12,958	$ 11,353	$ 12,084	$ 732	6.44%	$ 5,335	$ 6,067	$ 732	13.71%
Feb-24	24,434	$ 21,276	$ 22,652	$ 1,375	6.46%	$ 9,930	$ 11,306	$ 1,375	13.85%
Mar-24	44,791	$ 38,880	$ 41,397	$ 2,517	6.47%	$ 18,081	$ 20,598	$ 2,517	13.92%
Annual Total	251,131	$ 218,931	$ 233,075	$ 14,145	6.46%	$ 102,315	$ 116,460	$ 14,145	13.82%

Work papers to develop SC 21 Rate 3 - Rate Year 2 Annual Bill Impacts

Annual Bills - RY1

$ 25.72				10MW	$ 0.0320	Nov-Mar																	
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit	BPP	Non-Labor O&M	DCSM	Net Plant Rec	MCG	MFC	GRT	MTA	Delivery	Monthly Totals	Total						
30	Apr-20	140,016	$ 771.69	$ 44,140	$ 32,204	$ (1,960)	$ 1.42	$ -	$ -	$ -	$ 601,520	$ 20,917	$ 16,791	$ 1,586		$ 77,136	$ 715,971						
31	May-20	129,717	$ 797.41	$ 44,140	$ 29,835	$ (1,816)	$ 1.42	$ -	$ -	$ -	$ 557,275	$ 19,378	$ 15,636	$ 1,477		$ 74,880	$ 666,724						
30	Jun-20	155,729	$ 771.69	$ 44,140	$ 35,818	$ (2,180)	$ 1.42	$ -	$ -	$ -	$ 669,024	$ 23,264	$ 18,554	$ 1,752		$ 80,620	$ 791,146						
31	Jul-20	116,977	$ 797.41	$ 44,140	$ 26,905	$ (1,638)	$ 1.42	$ -	$ -	$ -	$ 502,543	$ 17,475	$ 14,207	$ 1,342		$ 72,055	$ 605,772						
31	Aug-20	165,015	$ 797.41	$ 44,140	$ 37,953	$ (2,310)	$ 1.42	$ -	$ -	$ -	$ 708,918	$ 24,652	$ 19,597	$ 1,851		$ 82,705	$ 835,599						
30	Sep-20	111,761	$ 771.69	$ 44,140	$ 25,705	$ (1,565)	$ 1.42	$ -	$ -	$ -	$ 480,134	$ 16,696	$ 13,621	$ 1,286		$ 70,873	$ 580,791						
31	Oct-20	67,335	$ 797.41	$ 44,140	$ 15,487	$ (943)	$ 1.42	$ -	$ -	$ -	$ 289,277	$ 10,059	$ 8,637	$ 816		$ 61,050	$ 368,271						
30	Nov-20	74,817	$ 771.69	$ 44,140	$ 23,941	$ (1,047)	$ 1.42	$ -	$ -	$ -	$ 321,420	$ 11,177	$ 9,638	$ 910		$ 69,593	$ 410,952						
31	Dec-20	82,299	$ 797.41	$ 44,140	$ 26,336	$ (1,152)	$ 1.42	$ -	$ -	$ -	$ 353,563	$ 12,295	$ 10,494	$ 991		$ 71,970	$ 447,465						
31	Jan-21	82,299	$ 797.41	$ 44,140	$ 26,336	$ (1,152)	$ 1.42	$ -	$ -	$ -	$ 353,563	$ 12,295	$ 10,494	$ 991		$ 71,970	$ 447,465						
28	Feb-21	82,299	$ 720.24	$ 44,140	$ 26,336	$ (1,152)	$ 1.42	$ -	$ -	$ -	$ 353,563	$ 12,295	$ 10,492	$ 991		$ 71,890	$ 447,386						
31	Mar-21	67,335	$ 797.41	$ 44,140	$ 21,547	$ (943)	$ 1.42	$ -	$ -	$ -	$ 289,277	$ 10,059	$ 8,783	$ 830		$ 67,270	$ 374,491						
															Annual Totals	$ 872,012	$ 6,692,033						

Annual Bills - RY2

$ 25.72				10MW	$ 0.0321	Nov-Mar																	
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit	BPP	Non-Labor O&M	DCSM	Net Plant Rec	MCG	MFC	GRT	MTA	Delivery	Monthly Totals	Total						
30	Apr-21	140,016	$ 771.69	$ 44,140	$ 32,204	$ (1,960)	$ 1.26	$ -	$ -	$ -	$ 610,054	$ 20,917	$ 16,996	$ 1,605		$ 77,136	$ 724,729						
31	May-21	129,717	$ 797.41	$ 44,140	$ 29,835	$ (1,816)	$ 1.26	$ -	$ -	$ -	$ 565,181	$ 19,378	$ 15,826	$ 1,495		$ 74,879	$ 674,838						
30	Jun-21	155,729	$ 771.69	$ 44,140	$ 35,818	$ (2,180)	$ 1.26	$ -	$ -	$ -	$ 678,516	$ 23,264	$ 18,783	$ 1,774		$ 80,620	$ 800,887						
31	Jul-21	116,977	$ 797.41	$ 44,140	$ 26,905	$ (1,638)	$ 1.26	$ -	$ -	$ -	$ 509,672	$ 17,475	$ 14,378	$ 1,358		$ 72,055	$ 613,090						
31	Aug-21	165,015	$ 797.41	$ 47,710	$ 38,118	$ -	$ 1.26	$ -	$ 5,122	$ -	$ 718,975	$ 21,417	$ 20,030	$ 1,892		$ 94,166	$ 854,063						
30	Sep-21	111,761	$ 771.69	$ 47,710	$ 25,817	$ -	$ 1.26	$ -	$ 3,469	$ -	$ 486,946	$ 14,505	$ 13,942	$ 1,317		$ 79,817	$ 594,479						
31	Oct-21	67,335	$ 797.41	$ 47,710	$ 15,554	$ -	$ 1.26	$ -	$ 2,090	$ -	$ 293,381	$ 8,739	$ 8,864	$ 837		$ 67,896	$ 377,975						
30	Nov-21	74,817	$ 771.69	$ 47,710	$ 24,016	$ -	$ 1.26	$ -	$ 2,322	$ -	$ 325,980	$ 9,710	$ 9,881	$ 933		$ 76,793	$ 421,326						
31	Dec-21	82,299	$ 797.41	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 10,682	$ 10,753	$ 1,016		$ 79,522	$ 458,511						
31	Jan-22	82,299	$ 797.41	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 10,682	$ 10,753	$ 1,016		$ 79,522	$ 458,511						
28	Feb-22	82,299	$ 720.24	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 10,682	$ 10,751	$ 1,015		$ 79,443	$ 458,432						
31	Mar-22	67,335	$ 797.41	$ 47,710	$ 21,615	$ -	$ 1.26	$ -	$ 2,090	$ -	$ 293,381	$ 8,739	$ 9,010	$ 851		$ 74,116	$ 384,195						
															Annual Totals	$ 935,966	$ 6,821,035						

Total portion of annual bill assessed volumetrically	$ 6,253,406
Total annual bill	$ 6,821,035
Percentage of bill assessed volumetrically	**92%**

Work papers to develop SC 3 - Rate Year 2 Annual Bill Impacts

Annual Bills - RY1

		$ 39.51	$ 0.4404	$ 0.3238																				
		3	997	1000	$ 1.42	$ 0.01610	$ -	$ -	$ (0.01250)	$ -	$ 0.01494	$ 0.42961	4.548%	2.407%	0.222%	0.222%								
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR	O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply					Delivery	Total	
January	52,200	$ 39.51	$ 439.08	$ 16,578.56	$ 1.42	$ 840.42	$ -	$ -	$ (652.50)	$ -	$ 779.82	$ 22,425.54	$ 784.37	$ 558.55	$ 40.03	$ 52.76						$ 18,071	$ 41,888	
February	41,968	$ 39.51	$ 439.08	$ 13,265.44	$ 1.42	$ 675.68	$ -	$ -	$ (524.60)	$ -	$ 626.96	$ 18,029.79	$ 632.01	$ 449.07	$ 32.25	$ 42.41						$ 14,561	$ 33,709	
March	29,404	$ 39.51	$ 439.08	$ 9,197.22	$ 1.42	$ 473.40	$ -	$ -	$ (367.55)	$ -	$ 439.27	$ 12,632.19	$ 444.93	$ 314.63	$ 22.71	$ 29.72						$ 10,251	$ 23,667	
April	16,965	$ 39.51	$ 439.08	$ 5,169.47	$ 1.42	$ 273.14	$ -	$ -	$ (212.06)	$ -	$ 253.44	$ 7,288.30	$ 259.72	$ 181.53	$ 13.25	$ 17.15						$ 5,984	$ 13,724	
May	8,549	$ 39.51	$ 439.08	$ 2,444.37	$ 1.42	$ 137.64	$ -	$ -	$ (106.86)	$ -	$ 127.71	$ 3,672.72	$ 134.40	$ 91.48	$ 6.86	$ 8.64						$ 3,096	$ 6,997	
June	5,206	$ 39.51	$ 439.08	$ 1,361.90	$ 1.42	$ 83.82	$ -	$ -	$ (65.08)	$ -	$ 77.77	$ 2,236.54	$ 84.62	$ 55.71	$ 4.32	$ 5.26						$ 1,950	$ 4,325	
July	4,932	$ 39.51	$ 439.08	$ 1,273.18	$ 1.42	$ 79.41	$ -	$ -	$ (61.65)	$ -	$ 73.68	$ 2,118.83	$ 80.54	$ 52.77	$ 4.11	$ 4.98						$ 1,856	$ 4,106	
August	4,820	$ 39.51	$ 439.08	$ 1,236.92	$ 1.42	$ 77.60	$ -	$ -	$ (60.25)	$ -	$ 72.01	$ 2,070.71	$ 78.87	$ 51.58	$ 4.03	$ 4.87						$ 1,817	$ 4,016	
September	4,904	$ 39.51	$ 439.08	$ 1,264.12	$ 1.42	$ 78.95	$ -	$ -	$ (61.30)	$ -	$ 73.26	$ 2,106.80	$ 80.13	$ 52.47	$ 4.09	$ 4.96						$ 1,846	$ 4,083	
October	12,958	$ 39.51	$ 439.08	$ 3,872.00	$ 1.42	$ 208.62	$ -	$ -	$ (161.98)	$ -	$ 193.58	$ 5,566.86	$ 200.05	$ 138.65	$ 10.21	$ 13.10						$ 4,609	$ 10,521	
November	24,434	$ 39.51	$ 439.08	$ 7,587.93	$ 1.42	$ 393.39	$ -	$ -	$ (305.43)	$ -	$ 365.02	$ 10,497.04	$ 370.93	$ 261.45	$ 18.93	$ 24.69						$ 8,546	$ 19,694	
December	44,791	$ 39.51	$ 439.08	$ 14,179.53	$ 1.42	$ 721.14	$ -	$ -	$ (559.89)	$ -	$ 669.13	$ 19,242.57	$ 674.05	$ 479.27	$ 34.40	$ 45.27						$ 15,529	$ 35,965	
Total		$ 474.12	$ 5,268.95	$ 77,430.62	$ 17.04	$ 4,043.21	$ -	$ -	$ (3,139.14)	$ -	$ 3,751.65	$ 107,887.89	$ 3,824.63	$ 2,687.16	$ 195.18	$ 253.81						$ 88,115	$ 202,695	

Annual Bills - RY2

		$ 39.51	$ 0.4553	$ 0.3561																				
		3	997	1000	$ 1.26	$ -	$ -	$ 0.00443	$ -	$ -	$ 0.01494	$ 0.43750	4.548%	2.407%	0.222%	0.222%								
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR	O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply					Delivery	Total	
January	52,200	$ 39.51	$ 453.93	$ 18,232.32	$ 1.26	$ -	$ -	$ 231.19	$ -	$ -	$ 779.82	$ 22,837.66	$ 862.22	$ 568.47	$ 44.00	$ 53.69						$ 19,864	$ 44,104	
February	41,968	$ 39.51	$ 453.93	$ 14,588.70	$ 1.26	$ -	$ -	$ 185.88	$ -	$ -	$ 626.96	$ 18,361.13	$ 694.45	$ 457.04	$ 35.44	$ 43.17						$ 15,999	$ 35,487	
March	29,404	$ 39.51	$ 453.93	$ 10,114.66	$ 1.26	$ -	$ -	$ 130.23	$ -	$ -	$ 439.27	$ 12,864.34	$ 488.44	$ 320.22	$ 24.93	$ 30.24						$ 11,253	$ 24,907	
April	16,965	$ 39.51	$ 453.93	$ 5,685.14	$ 1.26	$ -	$ -	$ 75.14	$ -	$ -	$ 253.44	$ 7,422.24	$ 284.48	$ 184.75	$ 14.52	$ 17.45						$ 6,554	$ 14,432	
May	8,549	$ 39.51	$ 453.93	$ 2,688.20	$ 1.26	$ -	$ -	$ 37.86	$ -	$ -	$ 127.71	$ 3,740.21	$ 146.48	$ 93.10	$ 7.48	$ 8.79						$ 3,375	$ 7,345	
June	5,206	$ 39.51	$ 453.93	$ 1,497.76	$ 1.26	$ -	$ -	$ 23.06	$ -	$ -	$ 77.77	$ 2,277.64	$ 91.67	$ 56.69	$ 4.68	$ 5.35						$ 2,112	$ 4,529	
July	4,932	$ 39.51	$ 453.93	$ 1,400.19	$ 1.26	$ -	$ -	$ 21.84	$ -	$ -	$ 73.68	$ 2,157.76	$ 87.17	$ 53.71	$ 4.45	$ 5.07						$ 2,008	$ 4,299	
August	4,820	$ 39.51	$ 453.93	$ 1,360.30	$ 1.26	$ -	$ -	$ 21.35	$ -	$ -	$ 72.01	$ 2,108.76	$ 85.34	$ 52.49	$ 4.35	$ 4.96						$ 1,966	$ 4,204	
September	4,904	$ 39.51	$ 453.93	$ 1,390.21	$ 1.26	$ -	$ -	$ 21.72	$ -	$ -	$ 73.26	$ 2,145.51	$ 86.71	$ 53.41	$ 4.43	$ 5.04						$ 1,998	$ 4,275	
October	12,958	$ 39.51	$ 453.93	$ 4,258.24	$ 1.26	$ -	$ -	$ 57.39	$ -	$ -	$ 193.58	$ 5,669.16	$ 218.77	$ 141.12	$ 11.16	$ 13.33						$ 5,040	$ 11,057	
November	24,434	$ 39.51	$ 453.93	$ 8,344.85	$ 1.26	$ -	$ -	$ 108.22	$ -	$ -	$ 365.02	$ 10,689.95	$ 406.94	$ 266.09	$ 20.77	$ 25.13						$ 9,375	$ 20,722	
December	44,791	$ 39.51	$ 453.93	$ 15,593.98	$ 1.26	$ -	$ -	$ 198.38	$ -	$ -	$ 669.13	$ 19,596.20	$ 740.74	$ 487.79	$ 37.80	$ 46.07						$ 17,066	$ 37,865	
Total		$ 474.12	$ 5,447.21	$ 85,154.55	$ 15.12	$ -	$ -	$ 1,112.26	$ -	$ -	$ 3,751.65	$ 109,870.57	$ 4,193.40	$ 2,734.89	$ 214.00	$ 258.31						$ 96,611	$ 213,226	

Work papers to develop SC 21 Rate 3 - Rate Year 3 Annual Bill Impacts

Annual Bills - RY2

10MW $ 0.0321 Nov-Mar

$ 25.72			$ 771.69	$ 4,771	$ 0.0231	$ -	$ 1.26	$ -	$ 0.0031	$ -	$ 0.4357	$ 0.0130	2.407%	0.222%		Monthly Totals	
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit	BPP	Non-Labor O&M	DCSM	Net Plant Rec	MCG	MFC	GRT	MTA		Delivery	Total
30	Apr-21	140,016	$ 771.69	$ 44,140	$ 32,204	$ (1,960)	$ 1.26	$ -	$ -	$ -	$ 610,054	$ 20,917	$ 16,996	$ 1,605		$ 77,136	$ 724,729
31	May-21	129,717	$ 797.41	$ 44,140	$ 29,835	$ (1,816)	$ 1.26	$ -	$ -	$ -	$ 565,181	$ 19,378	$ 15,826	$ 1,495		$ 74,879	$ 674,838
30	Jun-21	155,729	$ 771.69	$ 44,140	$ 35,818	$ (2,180)	$ 1.26	$ -	$ -	$ -	$ 678,516	$ 23,264	$ 18,783	$ 1,774		$ 80,620	$ 800,887
31	Jul-21	116,977	$ 797.41	$ 44,140	$ 26,905	$ (1,638)	$ 1.26	$ -	$ -	$ -	$ 509,672	$ 17,475	$ 14,378	$ 1,358		$ 72,055	$ 613,090
31	Aug-21	165,015	$ 797.41	$ 47,710	$ 38,118	$ -	$ 1.26	$ -	$ 5,122	$ -	$ 718,975	$ 24,652	$ 20,108	$ 1,899		$ 94,166	$ 857,383
30	Sep-21	111,761	$ 771.69	$ 47,710	$ 25,817	$ -	$ 1.26	$ -	$ 3,469	$ -	$ 486,946	$ 16,696	$ 13,995	$ 1,322		$ 79,817	$ 596,727
31	Oct-21	67,335	$ 797.41	$ 47,710	$ 15,554	$ -	$ 1.26	$ -	$ 2,090	$ -	$ 293,381	$ 10,059	$ 8,896	$ 840		$ 67,896	$ 379,329
30	Nov-21	74,817	$ 771.69	$ 47,710	$ 24,016	$ -	$ 1.26	$ -	$ 2,322	$ -	$ 325,980	$ 11,177	$ 9,916	$ 937		$ 76,793	$ 422,831
31	Dec-21	82,299	$ 797.41	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 12,295	$ 10,792	$ 1,019		$ 79,522	$ 460,166
31	Jan-22	82,299	$ 797.41	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 12,295	$ 10,792	$ 1,019		$ 79,522	$ 460,166
28	Feb-22	82,299	$ 720.24	$ 47,710	$ 26,418	$ -	$ 1.26	$ -	$ 2,555	$ -	$ 358,579	$ 12,295	$ 10,790	$ 1,019		$ 79,443	$ 460,087
31	Mar-22	67,335	$ 797.41	$ 47,710	$ 21,615	$ -	$ 1.26	$ -	$ 2,090	$ -	$ 293,381	$ 10,059	$ 9,042	$ 854	Annual Totals	$ 74,116	$ 385,549
																$ 935,966	$ 6,835,783

Annual Bills - RY3

10MW $ 0.0321 Nov-Mar

$ 25.72			$ 771.69	$ 4,928	$ 0.0230	$ -	$ 1.31	$ 0.0001	$ 0.0127	$ -	$ 0.4357	$ 0.0130	2.407%	0.222%		Monthly Totals	
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit	BPP	Non-Labor O&M	DCSM	Net Plant Rec	MCG	MFC	GRT	MTA		Delivery	Total
30	Apr-22	140,016	$ 771.69	$ 49,280	$ 32,204	$ -	$ 1.31	$ 143	$ 17,846	$ -	$ 610,054	$ 18,173	$ 17,534	$ 1,656		$ 102,887	$ 747,663
31	May-22	129,717	$ 797.41	$ 49,280	$ 29,835	$ -	$ 1.31	$ 132	$ 16,534	$ -	$ 565,181	$ 16,836	$ 16,334	$ 1,543		$ 99,124	$ 696,473
30	Jun-22	155,729	$ 771.69	$ 49,280	$ 35,818	$ -	$ 1.31	$ 159	$ 19,849	$ -	$ 678,516	$ 20,212	$ 19,367	$ 1,829		$ 108,668	$ 825,803
31	Jul-22	116,977	$ 797.41	$ 49,280	$ 26,905	$ -	$ 1.31	$ 119	$ 14,910	$ -	$ 509,672	$ 15,182	$ 14,848	$ 1,402		$ 94,437	$ 633,118
31	Aug-22	165,015	$ 797.41	$ 49,280	$ 37,953	$ -	$ 1.31	$ 168	$ 21,033	$ -	$ 718,975	$ 21,417	$ 20,450	$ 1,932		$ 112,111	$ 872,008
30	Sep-22	111,761	$ 771.69	$ 49,280	$ 25,705	$ -	$ 1.31	$ 114	$ 14,245	$ -	$ 486,946	$ 14,505	$ 14,239	$ 1,345		$ 92,491	$ 607,153
31	Oct-22	67,335	$ 797.41	$ 49,280	$ 15,487	$ -	$ 1.31	$ 69	$ 8,583	$ -	$ 293,381	$ 8,739	$ 9,058	$ 856		$ 76,172	$ 386,251
30	Nov-22	74,817	$ 771.69	$ 49,280	$ 24,016	$ -	$ 1.31	$ 76	$ 9,536	$ -	$ 325,980	$ 9,710	$ 10,094	$ 953		$ 85,886	$ 430,420
31	Dec-22	82,299	$ 797.41	$ 49,280	$ 26,418	$ -	$ 1.31	$ 84	$ 10,490	$ -	$ 358,579	$ 10,682	$ 10,984	$ 1,037		$ 89,364	$ 468,353
31	Jan-23	82,299	$ 797.41	$ 49,280	$ 26,418	$ -	$ 1.31	$ 84	$ 10,490	$ -	$ 358,579	$ 10,682	$ 10,984	$ 1,037		$ 89,364	$ 468,353
28	Feb-23	82,299	$ 720.24	$ 49,280	$ 26,418	$ -	$ 1.31	$ 84	$ 10,490	$ -	$ 358,579	$ 10,682	$ 10,982	$ 1,037		$ 89,285	$ 468,273
31	Mar-23	67,335	$ 797.41	$ 49,280	$ 21,615	$ -	$ 1.31	$ 69	$ 8,583	$ -	$ 293,381	$ 8,739	$ 9,206	$ 870	Annual Totals	$ 82,461	$ 392,540
																$ 1,122,250	$ 6,996,407

Total portion of annual bill assessed volumetrically	$ 6,395,658
Total annual bill	$ 6,996,407
Percentage of bill assessed volumetrically	**91%**

Work papers to develop SC 3 - Rate Year 3 Annual Bill Impacts

Annual Bills - RY2

		$ 39.51	$ 0.4553	$ 0.3561																					
		3	997	1000	$ 1.26	$ -	$ -	$ 0.00443	$ -	$ -	$ 0.01494	$ 0.43750	4.548%	2.407%	0.222%	0.222%									
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply	Delivery	Total							
January	52,200	$ 39.51	$ 453.93	$ 18,232.32	$ 1.26	$ -	$ -	$ 231.19	$ -	$ -	$ 779.82	$ 22,837.66	$ 862.22	$ 568.47	$ 44.00	$ 53.69	$ 19,864	$ 44,104							
February	41,968	$ 39.51	$ 453.93	$ 14,588.70	$ 1.26	$ -	$ -	$ 185.88	$ -	$ -	$ 626.96	$ 18,361.13	$ 694.45	$ 457.04	$ 35.44	$ 43.17	$ 15,999	$ 35,487							
March	29,404	$ 39.51	$ 453.93	$ 10,114.66	$ 1.26	$ -	$ -	$ 130.23	$ -	$ -	$ 439.27	$ 12,864.34	$ 488.44	$ 320.22	$ 24.93	$ 30.24	$ 11,253	$ 24,907							
April	16,965	$ 39.51	$ 453.93	$ 5,685.14	$ 1.26	$ -	$ -	$ 75.14	$ -	$ -	$ 253.44	$ 7,422.24	$ 284.48	$ 184.75	$ 14.52	$ 17.45	$ 6,554	$ 14,432							
May	8,549	$ 39.51	$ 453.93	$ 2,688.20	$ 1.26	$ -	$ -	$ 37.86	$ -	$ -	$ 127.71	$ 3,740.21	$ 146.48	$ 93.10	$ 7.48	$ 8.79	$ 3,375	$ 7,345							
June	5,206	$ 39.51	$ 453.93	$ 1,497.76	$ 1.26	$ -	$ -	$ 23.06	$ -	$ -	$ 77.77	$ 2,277.64	$ 91.67	$ 56.69	$ 4.68	$ 5.35	$ 2,112	$ 4,529							
July	4,932	$ 39.51	$ 453.93	$ 1,400.19	$ 1.26	$ -	$ -	$ 21.84	$ -	$ -	$ 73.68	$ 2,157.76	$ 87.17	$ 53.71	$ 4.45	$ 5.07	$ 2,008	$ 4,299							
August	4,820	$ 39.51	$ 453.93	$ 1,360.30	$ 1.26	$ -	$ -	$ 21.35	$ -	$ -	$ 72.01	$ 2,108.76	$ 85.34	$ 52.49	$ 4.35	$ 4.96	$ 1,966	$ 4,204							
September	4,904	$ 39.51	$ 453.93	$ 1,390.21	$ 1.26	$ -	$ -	$ 21.72	$ -	$ -	$ 73.26	$ 2,145.51	$ 86.71	$ 53.41	$ 4.43	$ 5.04	$ 1,998	$ 4,275							
October	12,958	$ 39.51	$ 453.93	$ 4,258.24	$ 1.26	$ -	$ -	$ 57.39	$ -	$ -	$ 193.58	$ 5,669.16	$ 218.77	$ 141.12	$ 11.16	$ 13.33	$ 5,040	$ 11,057							
November	24,434	$ 39.51	$ 453.93	$ 8,344.85	$ 1.26	$ -	$ -	$ 108.22	$ -	$ -	$ 365.02	$ 10,689.95	$ 406.94	$ 266.09	$ 20.77	$ 25.13	$ 9,375	$ 20,722							
December	44,791	$ 39.51	$ 453.93	$ 15,593.98	$ 1.26	$ -	$ -	$ 198.38	$ -	$ -	$ 669.13	$ 19,596.20	$ 740.74	$ 487.79	$ 37.80	$ 46.07	$ 17,066	$ 37,865							
Total		$ 474.12	$ 5,447.21	$ 85,154.55	$ 15.12	$ -	$ -	$ 1,112.26	$ -	$ -	$ 3,751.65	$ 109,870.57	$ 4,193.40	$ 2,734.89	$ 214.00	$ 258.31	$ 96,611	$ 213,226							

Annual Bills - RY3

		$ 39.51	$ 0.4638	$ 0.3628																					
		3	997	1000	$ 1.31	$ -	$ 0.00085	$ 0.01847	$ -	$ -	$ 0.01494	$ 0.43750	4.548%	2.407%	0.222%	0.222%									
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply	Delivery	Total							
January	52,200	$ 39.51	$ 462.41	$ 18,575.36	$ 1.31	$ -	$ 44.27	$ 964.29	$ -	$ -	$ 779.82	$ 22,837.66	$ 913.56	$ 568.47	$ 46.62	$ 53.69	$ 21,047	$ 45,287							
February	41,968	$ 39.51	$ 462.41	$ 14,863.19	$ 1.31	$ -	$ 35.59	$ 775.27	$ -	$ -	$ 626.96	$ 18,361.13	$ 735.74	$ 457.04	$ 37.55	$ 43.17	$ 16,951	$ 36,439							
March	29,404	$ 39.51	$ 462.41	$ 10,304.97	$ 1.31	$ -	$ 24.93	$ 543.18	$ -	$ -	$ 439.27	$ 12,864.34	$ 517.39	$ 320.22	$ 26.40	$ 30.24	$ 11,920	$ 25,574							
April	16,965	$ 39.51	$ 462.41	$ 5,792.10	$ 1.31	$ -	$ 14.39	$ 313.39	$ -	$ -	$ 253.44	$ 7,422.24	$ 301.22	$ 184.75	$ 15.37	$ 17.45	$ 6,940	$ 14,818							
May	8,549	$ 39.51	$ 462.41	$ 2,738.78	$ 1.31	$ -	$ 7.25	$ 157.93	$ -	$ -	$ 127.71	$ 3,740.21	$ 154.96	$ 93.10	$ 7.91	$ 8.79	$ 3,570	$ 7,540							
June	5,206	$ 39.51	$ 462.41	$ 1,525.94	$ 1.31	$ -	$ 4.41	$ 96.17	$ -	$ -	$ 77.77	$ 2,277.64	$ 96.86	$ 56.69	$ 4.94	$ 5.35	$ 2,232	$ 4,649							
July	4,932	$ 39.51	$ 462.41	$ 1,426.53	$ 1.31	$ -	$ 4.18	$ 91.11	$ -	$ -	$ 73.68	$ 2,157.76	$ 92.10	$ 53.71	$ 4.70	$ 5.07	$ 2,122	$ 4,412							
August	4,820	$ 39.51	$ 462.41	$ 1,385.90	$ 1.31	$ -	$ 4.09	$ 89.04	$ -	$ -	$ 72.01	$ 2,108.76	$ 90.15	$ 52.49	$ 4.60	$ 4.96	$ 2,077	$ 4,315							
September	4,904	$ 39.51	$ 462.41	$ 1,416.37	$ 1.31	$ -	$ 4.16	$ 90.59	$ -	$ -	$ 73.26	$ 2,145.51	$ 91.61	$ 53.41	$ 4.68	$ 5.04	$ 2,111	$ 4,388							
October	12,958	$ 39.51	$ 462.41	$ 4,338.36	$ 1.31	$ -	$ 10.99	$ 239.37	$ -	$ -	$ 193.58	$ 5,669.16	$ 231.58	$ 141.12	$ 11.82	$ 13.33	$ 5,335	$ 11,353							
November	24,434	$ 39.51	$ 462.41	$ 8,501.86	$ 1.31	$ -	$ 20.72	$ 451.37	$ -	$ -	$ 365.02	$ 10,689.95	$ 431.02	$ 266.09	$ 22.00	$ 25.13	$ 9,930	$ 21,276							
December	44,791	$ 39.51	$ 462.41	$ 15,887.37	$ 1.31	$ -	$ 37.98	$ 827.42	$ -	$ -	$ 669.13	$ 19,596.20	$ 784.80	$ 487.79	$ 40.05	$ 46.07	$ 18,081	$ 38,880							
Total		$ 474.12	$ 5,548.90	$ 86,756.73	$ 15.72	$ -	$ 212.96	$ 4,639.14	$ -	$ -	$ 3,751.65	$ 109,870.57	$ 4,441.01	$ 2,734.89	$ 226.64	$ 258.31	$ 102,315	$ 218,931							

Work papers to develop SC 21 Rate 3 - Rate Year 4 Annual Bill Impacts

Annual Bills - RY3

$ 25.72			$ 771.69	10MW	$ 4,928	$ 0.0321	Nov-Mar	$ -	$ 1.31	$ 0.0001		$ 0.0127	$ -		$ 0.4357	$ 0.0130		2.407%		0.222%						Monthly Totals					
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit		BPP	Non-Labor O&M		DCSM	Net Plant Rec		MCG	MFC	GRT	MTA								Delivery		Total				
30	Apr-22	140,016	$ 771.69	$ 49,280	$ 32,204	$ -		$ 1.31	$ 143		$ 17,846	$ -		$ 610,054	$ 18,173	$ 17,534	$ 1,656								$ 102,887		$ 747,663				
31	May-22	129,717	$ 797.41	$ 49,280	$ 29,835	$ -		$ 1.31	$ 132		$ 16,534	$ -		$ 565,181	$ 16,836	$ 16,334	$ 1,543								$ 99,124		$ 696,473				
30	Jun-22	155,729	$ 771.69	$ 49,280	$ 35,818	$ -		$ 1.31	$ 159		$ 19,849	$ -		$ 678,516	$ 20,212	$ 19,367	$ 1,829								$ 108,668		$ 825,803				
31	Jul-22	116,977	$ 797.41	$ 49,280	$ 26,905	$ -		$ 1.31	$ 119		$ 14,910	$ -		$ 509,672	$ 15,182	$ 14,848	$ 1,402								$ 94,437		$ 633,118				
31	Aug-22	165,015	$ 797.41	$ 49,280	$ 37,953	$ -		$ 1.31	$ 168		$ 21,033	$ -		$ 718,975	$ 21,417	$ 20,450	$ 1,932								$ 112,111		$ 872,008				
30	Sep-22	111,761	$ 771.69	$ 49,280	$ 25,705	$ -		$ 1.31	$ 114		$ 14,245	$ -		$ 486,946	$ 14,505	$ 14,239	$ 1,345								$ 92,491		$ 607,153				
31	Oct-22	67,335	$ 797.41	$ 49,280	$ 15,487	$ -		$ 1.31	$ 69		$ 8,583	$ -		$ 293,381	$ 8,739	$ 9,058	$ 856								$ 76,172		$ 386,251				
30	Nov-22	74,817	$ 771.69	$ 49,280	$ 24,016	$ -		$ 1.31	$ 76		$ 9,536	$ -		$ 325,980	$ 9,710	$ 10,094	$ 953								$ 85,886		$ 430,420				
31	Dec-22	82,299	$ 797.41	$ 49,280	$ 26,418	$ -		$ 1.31	$ 84		$ 10,490	$ -		$ 358,579	$ 10,682	$ 10,984	$ 1,037								$ 89,364		$ 468,353				
31	Jan-23	82,299	$ 797.41	$ 49,280	$ 26,418	$ -		$ 1.31	$ 84		$ 10,490	$ -		$ 358,579	$ 10,682	$ 10,984	$ 1,037								$ 89,364		$ 468,353				
28	Feb-23	82,299	$ 720.24	$ 49,280	$ 26,418	$ -		$ 1.31	$ 84		$ 10,490	$ -		$ 358,579	$ 10,682	$ 10,982	$ 1,037								$ 89,285		$ 468,273				
31	Mar-23	67,335	$ 797.41	$ 49,280	$ 21,615	$ -		$ 1.31	$ 69		$ 8,583	$ -		$ 293,381	$ 8,739	$ 9,206	$ 870								$ 82,461		$ 392,540				
																								Annual Totals		$ 1,122,250		$ 6,996,407		

Annual Bills - RY4

$ 25.72			$ 771.69	10MW	$ 4,928	$ 0.0358	Nov-Mar	$ -	$ 1.31	$ 0.0001		$ 0.0287	$ 0.0019		$ 0.4357	$ 0.0130		2.407%		0.222%						Monthly Totals					
	Month	Usage (Dth)	Min Charge	Demand	Usage	Tax Reform Surcredit		BPP	Non-Labor O&M		DCSM	Net Plant Rec		MCG	MFC	GRT	MTA								Delivery		Total				
30	Apr-23	140,016	$ 771.69	$ 49,280	$ 35,984	$ -		$ 1.31	$ 161		$ 40,124	$ 2,663		$ 610,054	$ 18,173	$ 18,226	$ 1,721								$ 132,384		$ 777,160				
31	May-23	129,717	$ 797.41	$ 49,280	$ 33,337	$ -		$ 1.31	$ 149		$ 37,173	$ 2,467		$ 565,181	$ 16,836	$ 16,975	$ 1,603								$ 126,451		$ 723,800				
30	Jun-23	155,729	$ 771.69	$ 49,280	$ 40,022	$ -		$ 1.31	$ 179		$ 44,627	$ 2,962		$ 678,516	$ 20,212	$ 20,136	$ 1,902								$ 141,475		$ 858,610				
31	Jul-23	116,977	$ 797.41	$ 49,280	$ 30,063	$ -		$ 1.31	$ 135		$ 33,522	$ 2,225		$ 509,672	$ 15,182	$ 15,426	$ 1,457								$ 119,080		$ 657,761				
31	Aug-23	165,015	$ 797.41	$ 49,280	$ 42,409	$ -		$ 1.31	$ 190		$ 47,288	$ 3,139		$ 718,975	$ 21,417	$ 21,266	$ 2,009								$ 146,874		$ 906,771				
30	Sep-23	111,761	$ 771.69	$ 49,280	$ 28,723	$ -		$ 1.31	$ 129		$ 32,027	$ 2,126		$ 486,946	$ 14,505	$ 14,791	$ 1,397								$ 116,035		$ 630,697				
31	Oct-23	67,335	$ 797.41	$ 49,280	$ 17,305	$ -		$ 1.31	$ 77		$ 19,296	$ 1,281		$ 293,381	$ 8,739	$ 9,391	$ 887								$ 90,357		$ 400,436				
30	Nov-23	74,817	$ 771.69	$ 49,280	$ 26,784	$ -		$ 1.31	$ 86		$ 21,440	$ 1,423		$ 325,980	$ 9,710	$ 10,482	$ 990								$ 102,416		$ 446,949				
31	Dec-23	82,299	$ 797.41	$ 49,280	$ 29,463	$ -		$ 1.31	$ 95		$ 23,584	$ 1,565		$ 358,579	$ 10,682	$ 11,410	$ 1,078								$ 107,547		$ 486,535				
31	Jan-24	82,299	$ 797.41	$ 49,280	$ 29,463	$ -		$ 1.31	$ 95		$ 23,584	$ 1,565		$ 358,579	$ 10,682	$ 11,410	$ 1,078								$ 107,547		$ 486,535				
28	Feb-24	82,299	$ 720.24	$ 49,280	$ 29,463	$ -		$ 1.31	$ 95		$ 23,584	$ 1,565		$ 358,579	$ 10,682	$ 11,408	$ 1,078								$ 107,467		$ 486,456				
31	Mar-24	67,335	$ 797.41	$ 49,280	$ 24,106	$ -		$ 1.31	$ 77		$ 19,296	$ 1,281		$ 293,381	$ 8,739	$ 9,555	$ 902								$ 97,337		$ 407,416				
																								Annual Totals		$ 1,394,970		$ 7,269,126		

Total portion of annual bill assessed volumetrically	$ 6,668,378
Total annual bill	$ 7,269,126
Percentage of bill assessed volumetrically	**92%**

Work papers to develop SC 3 - Rate Year 4 Annual Bill Impacts

Annual Bills - RY3

		$ 39.51	$ 0.4638	$ 0.3628																			
		3	997	1000	$ 1.31	$ -	$ 0.00085	$ 0.01847	$ -	$ -	$ 0.01494	$ 0.43750	4.548%	2.407%	0.222%	0.222%						
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply	Delivery	Total				
January	52,200	$ 39.51	$ 462.41	$ 18,575.36	$ 1.31	$ -	$ 44.27	$ 964.29	$ -	$ -	$ 779.82	$ 22,837.66	$ 913.56	$ 568.47	$ 46.62	$ 53.69	$ 21,047	$ 45,287				
February	41,968	$ 39.51	$ 462.41	$ 14,863.19	$ 1.31	$ -	$ 35.59	$ 775.27	$ -	$ -	$ 626.96	$ 18,361.13	$ 735.74	$ 457.04	$ 37.55	$ 43.17	$ 16,951	$ 36,439				
March	29,404	$ 39.51	$ 462.41	$ 10,304.97	$ 1.31	$ -	$ 24.93	$ 543.18	$ -	$ -	$ 439.27	$ 12,864.34	$ 517.39	$ 320.22	$ 26.40	$ 30.24	$ 11,920	$ 25,574				
April	16,965	$ 39.51	$ 462.41	$ 5,792.10	$ 1.31	$ -	$ 14.39	$ 313.39	$ -	$ -	$ 253.44	$ 7,422.24	$ 301.22	$ 184.75	$ 15.37	$ 17.45	$ 6,940	$ 14,818				
May	8,549	$ 39.51	$ 462.41	$ 2,738.78	$ 1.31	$ -	$ 7.25	$ 157.93	$ -	$ -	$ 127.71	$ 3,740.21	$ 154.96	$ 93.10	$ 7.91	$ 8.79	$ 3,570	$ 7,540				
June	5,206	$ 39.51	$ 462.41	$ 1,525.94	$ 1.31	$ -	$ 4.41	$ 96.17	$ -	$ -	$ 77.77	$ 2,277.64	$ 96.86	$ 56.69	$ 4.94	$ 5.35	$ 2,232	$ 4,649				
July	4,932	$ 39.51	$ 462.41	$ 1,426.53	$ 1.31	$ -	$ 4.18	$ 91.11	$ -	$ -	$ 73.68	$ 2,157.76	$ 92.10	$ 53.71	$ 4.70	$ 5.07	$ 2,122	$ 4,412				
August	4,820	$ 39.51	$ 462.41	$ 1,385.90	$ 1.31	$ -	$ 4.09	$ 89.04	$ -	$ -	$ 72.01	$ 2,108.76	$ 90.15	$ 52.49	$ 4.60	$ 4.96	$ 2,077	$ 4,315				
September	4,904	$ 39.51	$ 462.41	$ 1,416.37	$ 1.31	$ -	$ 4.16	$ 90.59	$ -	$ -	$ 73.26	$ 2,145.51	$ 91.61	$ 53.41	$ 4.68	$ 5.04	$ 2,111	$ 4,388				
October	12,958	$ 39.51	$ 462.41	$ 4,338.36	$ 1.31	$ -	$ 10.99	$ 239.37	$ -	$ -	$ 193.58	$ 5,669.16	$ 231.58	$ 141.12	$ 11.82	$ 13.33	$ 5,335	$ 11,353				
November	24,434	$ 39.51	$ 462.41	$ 8,501.86	$ 1.31	$ -	$ 20.72	$ 451.37	$ -	$ -	$ 365.02	$ 10,689.95	$ 431.02	$ 266.09	$ 22.00	$ 25.13	$ 9,930	$ 21,276				
December	44,791	$ 39.51	$ 462.41	$ 15,887.37	$ 1.31	$ -	$ 37.98	$ 827.42	$ -	$ -	$ 669.13	$ 19,596.20	$ 784.80	$ 487.79	$ 40.05	$ 46.07	$ 18,081	$ 38,880				
Total		$ 474.12	$ 5,548.90	$ 86,756.73	$ 15.72	$ -	$ 212.96	$ 4,639.14	$ -	$ -	$ 3,751.65	$ 109,870.57	$ 4,441.01	$ 2,734.89	$ 226.64	$ 258.31	$ 102,315	$ 218,931				

Annual Bills - RY4

		$ 39.51	$ 0.4849	$ 0.3793																			
		3	997	1000	$ 1.31	$ -	$ 0.00093	$ 0.03958	$ -	$ 0.01585	$ 0.01494	$ 0.43750	4.548%	2.407%	0.222%	0.222%						
Month	Usage (Therms)	First 3 Therms	Next 997	Over 1000	BPP	SBC	Non Labor DR O&M	DCSM	Tax Reform Surcredit	Net Plant Rec	MFC	MCG	GRT - Delivery	GRT - Supply	MTA - Delivery	MTA - Supply	Delivery	Total				
January	52,200	$ 39.51	$ 483.45	$ 19,420.16	$ 1.31	$ -	$ 48.65	$ 2,066.18	$ -	$ 827.11	$ 779.82	$ 22,837.66	$ 1,040.87	$ 568.47	$ 53.12	$ 53.69	$ 23,980	$ 48,220				
February	41,968	$ 39.51	$ 483.45	$ 15,539.16	$ 1.31	$ -	$ 39.11	$ 1,661.18	$ -	$ 664.98	$ 626.96	$ 18,361.13	$ 838.14	$ 457.04	$ 42.77	$ 43.17	$ 19,310	$ 38,798				
March	29,404	$ 39.51	$ 483.45	$ 10,773.64	$ 1.31	$ -	$ 27.40	$ 1,163.87	$ -	$ 465.91	$ 439.27	$ 12,864.34	$ 589.20	$ 320.22	$ 30.07	$ 30.24	$ 13,574	$ 27,228				
April	16,965	$ 39.51	$ 483.45	$ 6,055.52	$ 1.31	$ -	$ 15.81	$ 671.51	$ -	$ 268.81	$ 253.44	$ 7,422.24	$ 342.73	$ 184.75	$ 17.49	$ 17.45	$ 7,896	$ 15,774				
May	8,549	$ 39.51	$ 483.45	$ 2,863.34	$ 1.31	$ -	$ 7.97	$ 338.39	$ -	$ 135.46	$ 127.71	$ 3,740.21	$ 175.98	$ 93.10	$ 8.98	$ 8.79	$ 4,054	$ 8,024				
June	5,206	$ 39.51	$ 483.45	$ 1,595.34	$ 1.31	$ -	$ 4.85	$ 206.06	$ -	$ 82.49	$ 77.77	$ 2,277.64	$ 109.74	$ 56.69	$ 5.60	$ 5.35	$ 2,528	$ 4,946				
July	4,932	$ 39.51	$ 483.45	$ 1,491.41	$ 1.31	$ -	$ 4.60	$ 195.22	$ -	$ 78.15	$ 73.68	$ 2,157.76	$ 104.31	$ 53.71	$ 5.32	$ 5.07	$ 2,403	$ 4,694				
August	4,820	$ 39.51	$ 483.45	$ 1,448.93	$ 1.31	$ -	$ 4.49	$ 190.79	$ -	$ 76.37	$ 72.01	$ 2,108.76	$ 102.10	$ 52.49	$ 5.21	$ 4.96	$ 2,352	$ 4,590				
September	4,904	$ 39.51	$ 483.45	$ 1,480.79	$ 1.31	$ -	$ 4.57	$ 194.11	$ -	$ 77.70	$ 73.26	$ 2,145.51	$ 103.76	$ 53.41	$ 5.30	$ 5.04	$ 2,390	$ 4,668				
October	12,958	$ 39.51	$ 483.45	$ 4,535.67	$ 1.31	$ -	$ 12.08	$ 512.90	$ -	$ 205.32	$ 193.58	$ 5,669.16	$ 263.34	$ 141.12	$ 13.44	$ 13.33	$ 6,067	$ 12,084				
November	24,434	$ 39.51	$ 483.45	$ 8,888.52	$ 1.31	$ -	$ 22.77	$ 967.15	$ -	$ 387.16	$ 365.02	$ 10,689.95	$ 490.72	$ 266.09	$ 25.04	$ 25.13	$ 11,306	$ 22,652				
December	44,791	$ 39.51	$ 483.45	$ 16,609.93	$ 1.31	$ -	$ 41.75	$ 1,772.92	$ -	$ 709.71	$ 669.13	$ 19,596.20	$ 894.07	$ 487.79	$ 45.63	$ 46.07	$ 20,598	$ 41,397				
Total		$ 474.12	$ 5,801.34	$ 90,702.39	$ 15.72	$ -	$ 234.05	$ 9,940.27	$ -	$ 3,979.17	$ 3,751.65	$ 109,870.57	$ 5,054.97	$ 2,734.89	$ 257.97	$ 258.31	$ 116,460	$ 233,075				

STATE OF NEW YORK
DEPARTMENT OF PUBLIC SERVICE

Case 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

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STAFF REPLY STATEMENT

APPENDIX 2

Appendix 2 to the Staff Reply Statement contains National Weather Service historical records for Central Park during the month of February. The information was sourced on June 14, 2021 from: https://www.weather.gov/okx/CentralParkHistorical and https://www.weather.gov/media/okx/Climate/Almanacs/nyc/nycfeb.pdf

This data is unofficial and for informational purposes only.

For official certified climate data please contact the National Centers for Environmental Information (NCEI)

Page last updated: January 3, 2020

National Weather Service Forecast Office New York, NY
Central Park, New York, NY
February
Records: 1869 - Present
Temperature/Precipitation/Snowfall Normals: 1981-2010
Heating/Cooling Degree Day Normals: 1981-2010
Record year is the last year record was broken or tied

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Sun																															
Rise (EST)	7:06	7:05	7:04	7:03	7:02	7:01	7:00	6:59	6:57	6:56	6:55	6:54	6:53	6:51	6:50	6:49	6:47	6:46	6:45	6:43	6:42	6:40	6:39	6:37	6:36	6:35	6:33	6:32	6:31		
Set (EST)	5:14	5:15	5:16	5:17	5:19	5:20	5:21	5:22	5:24	5:25	5:26	5:27	5:28	5:30	5:31	5:32	5:33	5:34	5:36	5:37	5:38	5:39	5:40	5:42	5:43	5:44	5:45	5:46	5:46		
Temperature																															
Normal Max	39	39	39	40	40	40	40	40	40	41	41	41	41	41	42	42	42	42	42	43	43	43	43	44	44	44	44	45			
Normal Min	27	27	27	28	28	28	28	28	28	28	28	28	29	29	29	29	29	29	30	30	30	30	30	30	31	31	31	31			
Normal Mean	33	33	33	34	34	34	34	34	34	34	35	35	35	35	35	35	36	36	36	36	36	37	37	37	37	37	38	38			
Record High	67	59	64	68	70	68	54	62	63	61	65	62	64	63	73	71	67	68	66	69	78	69	70	75	75	65	72	67	69		
Record High Year	1989	1988	1991	1991	1991	2008	1938	2017	1990	2001	2009	2018	1951	1946	1949	1954	1976	1981	1997	1939	2018	1997	1985	1985	1930	1890	1997	1976	1880		
Record Low	-2	-3	0	0	-6	-4	1	-7	-15	-6	-2	-3	-1	-1	-8	1	-5	0	1	2	4	8	5	-1	1	7	5	5	4		
Record Low Year	1920	1881	1955	1918	1918	1895	1910	1934	1934	1899	1899	1914	1914	2016	1943	1888	1896	1979	1936	2015	1896	1963	1889	1873	1914	1990	1900	1934	1884		
Record Low Max	12	16	16	17	7	10	13	21	8	7	9	9	11	15	18	15	10	16	16	16	21	21	21	15	16	22	19	21	21		
Record Low Max Year	1971	1971	1917	1996	1886	1895	1895	1934	1934	1899	1899	1899	1899	2016	1875	2003	1958	1958	1903	1885	1968	1963	1907	1914	1934	2003	1920	1980	1980		
Record High Min	50	42	46	46	51	41	41	48	46	43	50	45	47	50	49	44	46	48	53	48	55	46	55	58	51	49	49	47	46		
Record High Min Year	1988	1952	2006	1991	1991	2008	1990	1965	2005	1999	1966	1966	1880	1949	1984	2002	2011	1981	2017	1939	2018	1996	1985	2017	1930	1976	1976	2017	1971		
Accumulated Normal Degree Days																															
Month (Heating)	32	64	96	127	158	189	220	251	282	313	343	373	403	433	463	493	522	551	580	609	638	666	694	722	750	778	805	832			
Season (Heating)	2718	2750	2782	2813	2844	2875	2906	2937	2968	2999	3029	3059	3089	3119	3149	3179	3208	3237	3266	3295	3324	3352	3380	3408	3436	3464	3491	3518			
Month (Cooling)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Season (Cooling)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Record Precipitation																															
Precipitation (Amount)	2.12	2.98	1.55	2.10	1.43	2.74	2.96	1.15	1.74	2.63	2.74	1.66	2.42	1.59	1.73	1.40	1.49	1.50	2.15	3.07	1.86	2.39	1.38	1.69	2.11	1.87	1.56	1.21	2.26		
Precipitation (Year)	1915	1973	1972	1961	2014	1896	1941	2013	1885	1970	1886	2006	1966	1914	1984	1885	2003	1887	1908	1898	1902	1878	1928	1977	1916	1929	2013	1910	1896		
Snowfall (Amount)	9.8	5.0	8.0	11.4	5.9	15.5	12.5	8.0	14.0	10.4	12.8	24.1	10.7	8.1	5.4	9.9	16.3	3.5	12.7	12.5	8.0	6.0	4.8	6.0	9.4	11.5	4.0	9.0	1.1		
Snowfall (Year)	1934	2015	2014	1961	1920	1978	1967	1870	1969	1926	1994	2006	1899	1914	1922	1996	2003	1928	1979	1921	1929	2008	1972	1907	2010	2010	1873	1949	1968		

Monthly Normals

Max Temperature	41.6°	Precipitation	3.09"
Min Temperature	28.9°	Snowfall	9.2"
Mean Temperature	35.3°		

Monthly Extremes

Warmest	42.0° – 2018	Wettest	6.87" – 1869
Coolest	19.9° - 1934	Driest	0.46" - 1895
		Snowiest	36.9" - 2010

STATE OF NEW YORK
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STAFF REPLY STATEMENT

APPENDIX 3

Appendix 3 to the Staff Reply Statement contains U.S. Census Bureau information regarding residential heating sources in New York State and specifically in the counties served by The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid. The information was sourced on June 14, 2021 from: https://data.census.gov/cedsci/table?q=New%20York%20State%20residential%20heating%20fuel&g=0500000US36047,36059,36081,36085,36103&tid=ACSDT1Y2019.B25040&hidePreview=true

HOUSE HEATING FUEL

United States™ Census Bureau

Note: The table shown may have been modified by user selections. Some information may be missing.

DATA NOTES	
TABLE ID:	B25040
SURVEY/PROGRAM:	American Community Survey
VINTAGE:	2019
DATASET:	ACSDT1Y2019
PRODUCT:	ACS 1-Year Estimates Detailed Tables
UNIVERSE:	Occupied housing units
FTP URL:	None
API URL:	https://api.census.gov/data/2019/acs/acs1
USER SELECTIONS	
GEOS	Suffolk County, New York; Nassau County, New York; New York; Lumbee (state) SDTSA, NC; Anahola (Residential) Hawaiian Home Land, HI; Kings County, New York; Queens County, New York; Richmond County, New York
EXCLUDED COLUMNS	None
APPLIED FILTERS	None
APPLIED SORTS	None
WEB ADDRESS	https://data.census.gov/cedsci/table?q=New%20York%20State%20residential%20heating%20fuel&g=0500000US36047,36059,36081,36085,36103&tid=ACSDT1Y2019.B25040&hidePreview=true
TABLE NOTES	Although the American Community Survey (ACS) produces population, demographic and housing unit estimates, it is the Census Bureau's Population Estimates Program that produces and disseminates the official estimates of the population for the nation, states, counties, cities, and towns and estimates of housing units for states and counties.
	Supporting documentation on code lists, subject definitions, data accuracy, and statistical testing can be found on the American Community Survey website in the Technical Documentation section. Sample size and data quality measures (including coverage rates, allocation rates, and response rates) can be found on the American Community Survey website in the Methodology section.
	Source: U.S. Census Bureau, 2019 American Community Survey 1-Year Estimates
	Data are based on a sample and are subject to sampling variability. The degree of uncertainty for an estimate arising from sampling variability is represented through the use of a margin of error. The value shown here is the 90 percent margin of error. The margin of error can be interpreted roughly as providing a 90 percent probability that the interval defined by the estimate minus the margin of error and the estimate plus the margin of error (the lower and upper confidence bounds) contains the true value. In addition to sampling variability, the ACS estimates are subject to nonsampling error (for a discussion of nonsampling variability, see ACS Technical Documentation). The effect of nonsampling error is not represented in these tables.
	The 2019 American Community Survey (ACS) data generally reflect the September 2018 Office of Management and Budget (OMB) delineations of metropolitan and micropolitan statistical areas. In certain instances the names, codes, and boundaries of the principal cities shown in ACS tables may differ from the OMB delineations due to differences in the effective dates of the geographic entities.
	Estimates of urban and rural populations, housing units, and characteristics reflect boundaries of urban areas defined based on Census 2010 data. As a result, data for urban and rural areas from the ACS do not necessarily reflect the results of ongoing urbanization.

	Explanation of Symbols: * An "**" entry in the margin of error column indicates that either no sample observations or too few sample observations were available to compute a standard error and thus the margin of error. A statistical test is not appropriate. * An "-" entry in the estimate column indicates that either no sample observations or too few sample observations were available to compute an estimate, or a ratio of medians cannot be calculated because one or both of the median estimates falls in the lowest interval or upper interval of an open-ended distribution, or the margin of error associated with a median was larger than the median itself. * An "-" following a median estimate means the median falls in the lowest interval of an open-ended distribution. * An "+" following a median estimate means the median falls in the upper interval of an open-ended distribution. * An "***" entry in the margin of error column indicates that the median falls in the lowest interval or upper interval of an open-ended distribution. A statistical test is not appropriate. * An "*****" entry in the margin of error column indicates that the estimate is controlled. A statistical test for sampling variability is not appropriate. * An "N" entry in the estimate and margin of error columns indicates that data for this geographic area cannot be displayed because the number of sample cases is too small. * An "(X)" means that the estimate is not applicable or not available.
COLUMN NOTES	None

	New York		Kings County, New York	
Label	**Estimate**	**Margin of Error**	**Estimate**	**Margin of Error**
Total:	7,446,812	±13,820	978,091	±5,532
Utility gas	4,519,999	±21,054	764,741	±9,069
Bottled, tank, or LP gas	319,576	±7,464	22,369	±1,941
Electricity	911,445	±14,527	90,490	±4,567
Fuel oil, kerosene, etc.	1,393,560	±17,245	70,759	±3,723
Coal or coke	16,959	±1,942	227	±217
Wood	114,011	±4,637	155	±158
Solar energy	8,697	±1,483	651	±316
Other fuel	79,096	±4,529	10,132	±1,326
No fuel used	83,469	±4,462	18,567	±2,182

	Nassau County, New York		Queens County, New York	
Label	**Estimate**	**Margin of Error**	**Estimate**	**Margin of Error**
Total:	449,798	±2,410	784,552	±4,746
Utility gas	275,854	±5,641	600,522	±7,734
Bottled, tank, or LP gas	8,208	±1,211	15,895	±1,610
Electricity	24,206	±2,378	53,155	±3,329
Fuel oil, kerosene, etc.	138,093	±4,942	93,753	±4,553
Coal or coke	0	±207	200	±125
Wood	592	±515	282	±190
Solar energy	505	±341	885	±508
Other fuel	1,247	±550	7,840	±1,453
No fuel used	1,093	±495	12,020	±1,612

	Richmond County, New York		Suffolk County, New York	
Label	**Estimate**	**Margin of Error**	**Estimate**	**Margin of Error**
Total:	166,297	±2,678	499,744	±4,104
Utility gas	147,718	±3,126	212,060	±5,963
Bottled, tank, or LP gas	1,850	±660	15,744	±2,346
Electricity	7,293	±1,747	36,130	±3,586
Fuel oil, kerosene, etc.	8,036	±1,323	228,673	±5,812
Coal or coke	0	±207	992	±661
Wood	71	±81	1,433	±526
Solar energy	296	±225	1,168	±500
Other fuel	212	±176	2,206	±869
No fuel used	821	±402	1,338	±715

STATE OF NEW YORK
DEPARTMENT OF PUBLIC SERVICE

Case 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

Case 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.

Case 18-M-0270 - Petition for Approval Pursuant to Public Service Law Section 113(2), of a Proposed Allocation of Certain Tax Refunds between KeySpan Gas East Corporation d/b/a National Grid and Ratepayers.

STAFF REPLY STATEMENT

APPENDIX 4

Appendix 4 to the Staff Reply Statement contains the response of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid to information request SANE-26.

Date of Request: January 29, 2021 Request No. SANE-26
Due Date: February 8, 2021 NG Request No. NG-1511

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
Case Nos. 19-G-0309 & 19-G-0310
Gas Utilities Rates

Request for Information

FROM: Kim Fraczek

TO: National Grid

SUBJECT: Gas Infrastructure

Request:

1. Provide the total capital expenditures the Companies spent in calendar year 2020 that they see to recover through this case. Please indicate now much of the total budgeted cost of each of these projects has been spent in 2020, and how much is outstanding. Please break out expenditures for the following projects/items:

    a. Total
    b. Metropolitan Reliability Infrastructure Project
    c. Greenpoint Vaporization Expansion (13 & 14)
    d. Greenpoint CNG Injection Heaters
    e. Greenpoint CNG Trucking Capabilities
    f. LNG Truck Load/Unload Station
    g. Staten Island CNG Fill Station
    h. Vaporizers 7 & 8 Replacement
    i. Glenwood CNG Injection Facility
    j. Barrett CNG Injection Facility
    k. Inwood CNG Injection Facility
    l. Executive salaries

2. Please provide a status update on the following projects, including any permits and variances requested by project name and permitting authority, and including dates of application, issue, and expiration. If permits have been requested but not yet issued or rejected, please include that information as well. Or each project, please also indicate all additional permits and variances that the Companies intend to seek in the future (i.e.permits and variances that are pending application).

    a. Metropolitan Reliability Infrastructure Project
    b. Greenpoint Vaporization Expansion (13 & 14)
    c. Greenpoint CNG Injection Heaters

d. Greenpoint CNG Trucking Capabilities
e. LNG Truck Load/Unload Station
f. Staten Island CNG Fill Station
g. Vaporizers 7 & 8 Replacement
h. Glenwood CNG Injection Facility
i. Barrett CNG Injection Facility
j. Inwood CNG Injection Facility

Response:

1. Attachment 1 is a copy of the Companies' quarterly capital investment report, as of September 30, 2020, as filed with the Commission in Cases 16-G-0058 and 16-G-0059. The year-end capital report required by the Joint Proposal in those cases is due no later than March 1, 2021. The Companies will supplement this response with the year-end capital report once filed with the Commission.

2.

a. Metropolitan Reliability Infrastructure Project – Phase IV is Complete. Phase V is on hold. Please see the Companies' response to NONBKPIPELINE-1 for the associated permits and statuses.
b. Greenpoint Vaporizer 13 & 14 –The project is in progress. Please see the Companies' response to SANE-16 for the associated permits and statuses.
c. Greenpoint CNG Injection Heaters – This project is a non-pipeline alternative to the Greenpoint Vaporizer 13 & 14 project. The project is currently on hold. Please see the Companies' response to SANE-18 for the associated permits and statuses.
d. Greenpoint CNG Trucking Capabilities – This project is on hold pending the CNG Injection Heaters Project. The Companies' are communicating and collaborating on transportation plans and response plans with the FDNY.
e. LNG Truck Load/Unload Station – The project is in progress. Please see the Companies' response to SANE-16 for the associated permits and statuses. Please note the DOB Permit – Building (Kiosk) that was "Rejected" has been resubmitted and is under review. The DOB Permit – mechanical - AC that was "Pending" has been approved.
f. Staten Island CNG Fill Station – This project has been cancelled.
g. Vaporizers 7 & 8 Replacement – Project development phase has not started. No permits have been requested at this time.
h. Glenwood CNG Injection Facility – The project is complete and in-service. Please see the Companies' response to NONBKPIPELINE-1 for the associated permits and statuses. Please note the SWPPP NOI and Building Permit have been obtained. The Company recently obtained Nassau County Fire Marshal - Site Plan & Fire Alarm Installers approvals.
i. Barrett CNG Injection Facility – The project is in construction. Please see the Companies' response to NONBKPIPELINE-1 for the associated permits and

statuses. Additionally, the Nassau County Fire Marshal - Site Plan & Fire Alarm Installers applications are pending submission.

j. Inwood CNG Injection Facility – The project is complete and in-service. Please see the Companies' response to NONBKPIPELINE-1 for the associated permits and statuses. The Company also recently obtained the Nassau County Fire Marshal - Site Plan & Fire Alarm Installers approvals.

nationalgrid

Kara J. Krueger, Esq.
Senior Counsel I
New York Regulatory

November 10, 2020

VIA ELECTRONIC FILING

Honorable Michelle L. Phillips, Secretary
New York State Department of Public Service
3 Empire State Plaza, 19th Floor
Albany, NY 12223-1350

RE: Case 16-G-0058 – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service

Case 16-G-0059 – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service

Quarterly Capital Report for the Third Quarter of Calendar Year 2020

Dear Secretary Phillips:

Pursuant to the Commission's Order dated December 16, 2016 in the above-captioned matters, KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") and The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") hereby submit the Quarterly Capital Report for the third quarter of calendar year 2020. The quarterly variance reports and the leak prone pipe retirement progress reports are attached. The Type 3 leak repair and inventory summaries are as follows:

Type 3 Leaks as of September 30, 2020

	KEDLI	KEDNY
Type 3 leaks Repaired	176	95
Type 3 Inventory	6,481	1,800

Cumulative Type 3 leak repairs year to date

New Customer Connections as of September 30, 2020

	KEDLI	KEDNY
New Services Connected	2,997	1,217
New Accounts Established	3,597	2,771

300 Erie Boulevard West, Syracuse, New York 13202
T: 315-428-6611 ■ F: 315-401-7891 ■ kara.krueger@nationalgrid.com ■ www.nationalgrid.com

The following additional information is provided per the request of Department of Public Service Staff:

LPP	KEDNY	KEDLI
Total miles of LPP by the end of CY 2020	65	155
Reported CY 2020 Q3	28.8	62.5
Remaining to target	36.2	92.5

KEDNY

Metric	Min/Max	CY 2020 Target	Result as of 9/30/2020
Workable Leaks	Maximum	25	68
Total Leaks	Maximum	3,100	1,868

KEDLI

Metric	Min/Max	CY 2020 Target	Result as of 9/30/2020
Workable Leaks	Maximum	25	36
Total Leaks	Maximum	7,750	6,517

Please contact me with any questions regarding this filing. Thank you for your time and attention to this matter.

Sincerely,

/s/ Kara J. Krueger

Kara J. Krueger
National Grid
300 Erie Blvd. West, A4
Syracuse, New York 13202

Enc.

cc: Active Parties in Cases 16-G-0058 and 16-G-0059 (via DMM)
Nicola Jones, DPS Staff (via e-mail)
Mary Ann Sorrentino, DPS Staff (via e-mail)

KS GAS EAST CORP-KEDLI

CAPITAL BY CATEGORY - SEP CY2020 ($000's)

Classification	Category	Final Category Code	CY20 YTD - Actuals	CY20 YTD - Budget	CY20 YTD Variance to Budget	Spend Explanation
Growth	Base Growth	Automatic Meter Reading (AMR) - Growth	$0	$508	$508	Actuals charged to Non Infrastructure. AMR Reolacement
		Base Growth - Customer Contributions	($2,525)	($2,218)	$307	More credits than planned
		Base Growth - Install Meter / Regulator	$1,156	$438	($718)	More units than planned
		Base Growth - Meter Purchase / Operations	$0	$762	$762	Actuals charged to Mandated, Meter Purchases
		Build it Back Program	$12	($15)	($27)	Timing of Actuals - carryover of work from prior period
		Install Main	$9,428	$10,662	$1,235	Fewer units than planned
		Install Service	$18,289	$13,036	($5,253)	Higher unit cost
		Install Services Bare Main Replacement Program	$0	($14)	($14)	Minor variance not attributable to a specific cause
		NEP - Install Main	$1,515	$3,329	$1,814	Fewer units than planned
		NEP - Install Service	$501	$475	($26)	More units than planned
	Base Growth - Special Projects	AVALON BAY HUNTINGTON STATION	$36	($1)	($37)	Unplanned work
		Bloom Energy Fuel Cell Project	$499	$596	$97	Less work than planned
		THE MEADOWS AT YAPHANK	$0	$0	$0	
		UNITED RIVERHEAD TERMINAL	$0	$0	$0	
	Gas System Reinforcement	Gas Reinforcement - Flanders Phase I Transmisison main	$0	($2)	($2)	Minor variance not attributable to a specific cause
		Gas Reinforcement - Freeport Regulator Station	$953	$1,872	$919	Timing of Actuals - work completed in prior period
		Gas Reinforcement - Gas Reinforcement - LTLI12011 - Elmont - NL-25 Rebuild	$0	$8	$8	Budget reallocated to Gas Reinforcement - LTLI12011 - Elmont - NL-25 Rebuild
		Gas Reinforcement - Gas Reinforcement - LTLI12238 - Bellmore - Transmission	$0	$32	$32	Budget reallocated to Gas Reinforcement - LTLI12238 - Bellmore - Transmission
		Gas Reinforcement - LTLI10940 Wainscott - Industrial Av	($35)	($954)	($919)	Fewer credits than planned
		Gas Reinforcement - LTLI10985 SE Suffolk Infrastructure	$1,591	$1,486	($105)	More work than planned
		Gas Reinforcement - LTLI11874-NL-26 Rebuild Garden City	$0	$0	$0	
		Gas Reinforcement - LTLI11976 Amityville SL63 Rebuild	$49	$414	$365	Timing of Actuals - work completed in prior period
		Gas Reinforcement - LTLI12011 - Elmont - NL-25 Rebuild	$148	$360	$212	Budget reallocated from "Gas Reinforcement - Gas Reinforcement - LTLI12011 - Elmont - NL-25 Rebuild"
		Gas Reinforcement - LTLI12230 - Uniondale - New Main	$0	($0)	($0)	
		Gas Reinforcement - LTLI12238 - Bellmore - Transmission	$179	$490	$311	Less Work than planned. Budget reallocated from "Gas Reinforcement - Gas Reinforcement - LTLI12238 - Bellmore - Transmission"
		Gas Reinforcement - Merrick Reinforcement - New Main	$1,071	$453	($618)	Timing of Actuals - carryover of work from prior period
		Gas Reinforcement - Northport Regualtor Station	($2)	($55)	($53)	Fewer credits than planned
		Gas Reinforcement - SL-40 Rebuild Bayshore	($74)	$54	$129	Timing of Actuals - work completed in prior period
		Gas Reinforcement - Southampton Reinf Phase II	$0	($1)	($1)	Budget reallocated to "Gas Reinforcement - LTLI10941 - Southamoton Transmissio"
		Gas Reinforcement - LTLI10941 - Southampton Transmissio	$3	$0	($3)	Budget reallocated from "Gas Reinforcement - LTLI10941 - Southampton Transmissio"
		Gas Reinforcement - LTLI12622 Hampton BaysSL-19 Rebuild	$90	$55	($35)	Budget in "Pressure Regulating Facilities - SL-19" and "Gas Reinforcement - LTLI12622 Hampton BaysSL-19 Rebuild", to be combined. Less Work than planned.
		Fuel Cell Energy- Yaphank, William Floyd, Brookhaven	$0	$2,188	$2,188	Budget reallocated to "Fuel Center Energy-Yaphank."
		Fuel Cell Energy- Yaphank, William Floyd, Brookhaven (CIAC)	$0	($2,240)	($2,240)	Budget reallocated to "Fuel Center Energy-Yaphank."
		Reinforcement	$5,988	$9,868	$3,880	Work suspended due to COVID-19
	Special Projects	Fuel Center Energy-Yaphank.	($360)	$0	$360	More Credits than planned. Total Net Variance is a credit of $308K, which is a customer reimbursed project. Budget to be reallocated from "Fuel Cell Energy- Yaphank, William Floyd, Brookhaven (CIAC) and Fuel Cell Energy- Yaphank, William Floyd, Brookhaven "
Growth			**$38,511**	**$41,587**	**$3,076**	
Mandated	City State Construction - Non-Reimbursable	CSC-Non-Reimbursables	$2,181	$2,964	$784	Fewer units than planned
	City State Construction - Reimbursable	CSC-Reimbursable	$11,675	$2,998	($8,677)	More Units than planned
		CSC-Reimbursable - Farmingdale Realignment	$0	$0	$0	
	City State Construction - Reimbursements	CSC-Reimbursements	($1,443)	($474)	$969	More credits than planned
	Integrity	Corrosion	$525	$780	$255	Fewer units than planned
		Cross Bore Remediation	$492	$750	$258	Fewer units than planned
		GM-2 ILI Modifications	$0	($196)	($196)	Actuals charged to IMP-GM2-ILI mods Ph1
		GM-8; 16; 10 ICDA Drip Mods	$0	$201	$201	Other - Project deferred
		IMP GM-16/8 Emergent Pipe Replaceme	$3	($1)	($5)	Minor variance not attributable to a specific cause
		IMP-GM2-ILI Mods Ph1	$532	$0	($532)	Actuals charged from GM-2 ILI Modifications
		IMP-GM2-ILI Mods Ph2	$1,282	$0	($1,282)	Actuals charged and budget to be reallocated from IMP-GM2-Robotic ILI Mods; More work than planned
		IMP-GM2-Robotic ILI mods	$0	$617	$617	Actuals charged to IMP-GM2-ILI Mods Ph2. Budget to be reallocated to "IMP-GM2-ILI Mods Ph2"
		IMP GM-8 Drip 8 Removal and Pipe Re	($0)	($1)	($1)	Minor variance not attributable to a specific cause
		Meter Pitts	$0	$367	$367	FY21 Actuals charged to Meter Vault Remediation. Quartr 1 Budget for Meter Pits ($367) was for FY20 Only.
		NWN-Lake Success Phase 3A	$596	$372	($224)	More work than planned
		NY Joint Facilities	$6	($4)	($10)	Minor variance not attributable to a specific cause
		Pipeline Integrity-LI	$0	$83	$83	Fewer units than planned
		Transmission Station Integrity	$0	$923	$923	Fewer units than planned
		Operator Qualification Program	$0	$250	$250	Fewer units than planned
		Pipeline Integrity - IMP - 2/3 Drip Pot Modifications	$0	$96	$96	Fewer units than planned
		Pipeline Integrity - IMP - GM 10 Robotic ILI	$0	$192	$192	Fewer units than planned
		Pipeline Integrity - IMP-GM2-ILI Mods Ph1	$0	$3	$3	Fewer units than planned
		Pipeline Integrity - IMP - GM 2 Robotic ILI	$0	$2,425	$2,425	Work suspended due to COVID-19
		Pipeline Integrity - IVP	$0	$96	$96	Fewer units than planned
		Pipeline Integrity - IVP Reactive Main Replacement	$0	$192	$192	Fewer units than planned
		Restrictions for Elevated Gas Infrastructure	$0	$139	$139	Fewer units than planned
	Large Diameter CI Rehab	99# Broadway Hewlett Lining	$0	$0	$0	
		Cast Iron Main Lining-LI	$13	$0	($13)	Unplanned work
		Lining Project - E Rockaway Rd	$43	$61	$18	Timing of Actuals - work completed in prior period
		Lining Project - Rocklyn Ave	$0	($0)	($0)	Minor variance not attributable to a specific cause
		Lining Project - Waverly Ave	$19	$47	$28	Timing of Actuals - work completed in prior period
	Local Law 30	LL30	$0	($0)	($0)	Minor variance not attributable to a specific cause
	Main Replacements - Proactive Programs	Franklin Ave - Valley Stream	$506	$35	($471)	Timing of Actuals - carryover of work from prior period
		Main Repl Leak Prone Pipe	$169,828	$142,863	($26,965)	Higher unit cost, crews working premium time in attempt to hit CY/FY retirement targets
	Main Replacements - Reactive Programs	Maintenance	$2,249	$1,029	($1,220)	More units than planned
	Meter Work	Meter Changes	$1,511	$1,446	($64)	More Units than planned
		Other Meter Work	$9	($1)	($9)	Minor variance not attributable to a specific cause
	Purchase Meters (Replacements)	Meter Purchases	$3,771	$1,201	($2,570)	Actuals include Growth, Meter Purchases and more units than planned
	Service Replacements - Proactive Programs	Svc Repl-BS HP Leak Prone Services	($3)	($7)	($4)	Minor variance not attributable to a specific cause
	Service Replacements - Reactive Programs	Atmospheric Corrosion/Inside Inspections	$445	$200	($245)	More Units than planned
		Latent Damage Verification	$0	$316	$316	Fewer units than planned
		Leaks	$938	$1,234	$296	Fewer units than planned
		Non-Leak / Other	$1,685	$2,500	$815	Fewer units than planned
		Valve Installation/Replacement	$0	$73	$73	Fewer units than planned
	Special Projects	Access Protection Remediation-LI	$0	$159	$159	Actuals charged to "Access Protection Remediation-LI."
		Access Protection Remediation-LI.	$61	$0	($61)	Budget under "Access Protection Remediation-LI" and less work than planned
		Meter Vault Remediation	($1)	$322	$324	Less work than Planned in FY21. FY21 Actuals charged from Meter Pits
Mandated			**$196,921**	**$164,249**	**($32,672)**	
Reliability	CNG Programs	CNG - KEDLI Blanket	$0	$358	$358	Less work than planned
	CNG Programs - Special Projects	Hewlett CNG Rebuild	$1,596	$191	($1,406)	Timing of Actuals - carryover of work from prior period

KS GAS EAST CORP-KEDLI

CAPITAL BY CATEGORY - SEP CY2020 ($000's)

Classification	Category	Final Category Code	CY20 YTD - Actuals	CY20 YTD - Budget	CY20 YTD Variance to Budget	Spend Explanation
	Gas System Reliability	CNG - NY Brentwood - New Compressor	$0	$1,262	$1,262	Less work than planned
		EFB Gas Condensate Tk Retirement	$0	($3)	($3)	Minor variance not attributable to a specific cause
		Gas Planning/RCV Program	$673	$1,119	$446	Less work than planned
		Gas System Control	$0	$52	$52	Less work than planned
		Gas System Reliability - Gas Control (Training Simulator)	$0	$20	$20	Less work than planned
		GRS Heater Program-LI	$49	$435	$386	Less work than planned
		I&R - Reactive	$0	$166	$166	Less work than planned
		KEDLI Heaters - NPT Replacement	$0	$0	$0	
		LTLI11032-Westbury- RCV 023123400	$0	$8	$8	Minor variance not attributable to a specific cause
		LTLI11715- Westbury- RCV 023123413	$0	$28	$28	Minor variance not attributable to a specific cause
		NO-37 Merrick Refurbish	($63)	$950	$1,012	Other - Project Deferred, Capital Actuals reversed to O&M dollars
		NL-05	$7	$0	($7)	Unplanned work
		NL-07 Bellmore Refurbish	($68)	($9)	$59	Other - Project Deferred, Capital Actuals reversed to O&M dollars
		NL-23 Refurbishment	$563	$2,304	$1,740	Timing of Actuals - Deferred work
		PRE-SL-51 Flanders	$0	$0	$0	
		Pres Reg Facil - proactive-LI	$3	$2,623	$2,629	Less work than planned
		Pres Reg Facil - Reactive-LI	$417	$25	($392)	More work than planned
		Pressure Regulating Facilities - SL 14	$0	$33	$33	Less work than planned
		SL-52 Flanders Overhaul reg stat	$0	$105	$105	Timing of Actuals - work completed in prior period
		SL-53 Upgrade outlet piping	($10)	($66)	($56)	Timing of Actuals - work completed in prior period
		SL-54 Hauppauge Refurbish	$3,081	$2,713	($368)	Timing of Actuals - carryover of work from prior period
		Storm Hardening Install remote Service Shutoff Valves	$365	$2,915	$2,550	Timing of Actuals - Deferred work
		System Automation	$143	$622	$479	Fewer units than planned
		Water Intrusion	($973)	$130	$1,102	Budget reallocated to "Maintenance"
		LTLI12020- Deer Park- RCV 040632167-PM	$0	$10	$10	Minor variance not attributable to a specific cause
		LTLI12021- Deer Park- RCV 040632133-PM	$0	$10	$10	Minor variance not attributable to a specific cause
		LTLI12022- Pinelawn- RCV 041025722-PM	$0	$10	$10	Minor variance not attributable to a specific cause
		Pressure Regulating Facilities - SL-32	$0	$21	$21	Actuals charged to "SL-32 Refurbishment"
		LNG - Emergency Generator Upgrade	$0	$3	$3	Minor variance not attributable to a specific cause
		LNG - Piping Insulation Inspection & Upgrade	$0	$37	$37	Minor variance not attributable to a specific cause
		LNG - Security System Upgrade	$0	$192	$192	Less work than planned
		RNG - Renewable Natural Gas (RNG) Interconnections	$0	$173	$173	Less work than planned
		SL-32 Refurbishment	$34	$40	$6	Less work than planned. Budgeted under "Pressure Regulating Facilities - SL-32" and " SL-32 Refurbishment"
	LNG Programs	LNG - Blanket	$0	$764	$764	Less work than planned
		LNG - Holtsville LNG	$120	$19	($101)	More work than planned
		LNG - Holtsville (Process Safety).	$0	($0)	($0)	Minor variance not attributable to a specific cause
		MicroCHP Demonstration	($182)	($1)	$181	Prior period adjustment
		LNG-Piping Ins. Upgrade-KED	$71	$151	$80	Less work than planned
	LNG Programs - Special Projects	AESD System	$1,345	$2,532	$1,187	Less work than planned
		Controls System Upgrade	$4,067	$4,098	$31	Minor variance not attributable to a specific cause
		LNG - Boiloff Compressor System	$0	$299	$299	Less work than planned
		LNG - Hi Ex Foam System	$23	$149	$126	Less work than planned
		LNG - Liquefaction Critical Spares	$1,442	$349	($1,093)	Timing of Actuals - Invoices
		LNG - ReGen Heater Replacement	$0	$17	$17	Minor variance not attributable to a specific cause
		LNG - Tank Stair Replacement	$33	$47	$14	Minor variance not attributable to a specific cause
		LNG - Tank Upgrade	$335	$986	$651	Less work than planned
		P-20 Pump Upgrade	$0	$0	$0	
		Storage Building	$1,835	$2,788	$954	Less work than planned
		LNG - Tank FoundationHeating Systems Upgrade	$0	$38	$38	Minor variance not attributable to a specific cause
	LNG Special Projects	LNG - Analyzer Replacement 1	$0	$503	$503	Less work than planned
		LNG - Cyber Security Enhancements	$0	$225	$225	Less work than planned
		LNG - Hydrant System Piping Refurbishment	$0	$120	$120	Less work than planned
		LNG - Raw Gas Makeup System Replacement	$0	$33	$33	Less work than planned
		Power Center Upgrade	$0	$79	$79	Less work than planned
		Barrett CNG Off Loading Facility	$1,344	$4,334	$2,990	Less work than planned
		Bayshore Gate Station	$56	$114	$58	Less work than planned
		CNG - Contract Closeout	$0	$153	$153	Less work than planned
		Distribution Station Over Pressure Protection	$0	$502	$502	Less work than planned
	Special Projects	Breezy Pt - LP NorthEast	$0	$0	$0	
		Buried Vent Line Remediation KEDLI	$8	$92	$84	Less work than planned
		Gas Control Telemetry Upgrade 3G/4G	$2	$45	$43	Less work than planned
		Gas System Control - LI	$0	$45	$45	Less work than planned
		KEDLI CNG Facility Site 3	$3,576	$0	($3,576)	Unplanned work
		Long Beach M&R Refurbishment	$896	$1,018	$122	Timing of Actuals - work completed in prior period
		LTLI10651 RCV N/O QL04 Lynbrook	$152	$285	$133	Timing of Actuals-Design and/or material charges for future construction project
		LTLI11032-Westbury- RCV 023123410	$43	$70	$27	Timing of Actuals-Design and/or material charges for future construction project
		LTLI11985-Farmingdale-RCV 032583255	$182	$78	($104)	Timing of Actuals-Design and/or material charges for future construction project
		M2M Upgrade	$0	$42	$42	Less work than planned
		Northport M&R Station Refurbishment	$31	($51)	($82)	Timing of Actuals - carryover of work from prior period
		NWN Down Rate Gas Mains North of LI	$19	($3)	($22)	Timing of Actuals - carryover of work from prior period
		NWN Down Rate Gas Main South of LIE	$0	$0	$0	
		NWN Install two 350 to 233psi Reg S	$564	$1,397	$833	Timing of Actuals - work completed in prior period
		NWN LIE Interconnect	($0)	$0	$0	
		NWN N-S Interconnect Phase 2	$10,968	$12,918	$1,950	Work suspended due to COVID-19
		Riverhead CNG Project	$770	$574	($196)	More units than planned
		Rockville Centre M&R	$814	$193	($620)	More units than planned
		S Commack M&R Refurbish	($89)	$585	$674	Other - Tax Reversal
		Glenwood CNG Expansion	$2,349	$3,850	$1,501	Less work than planned
		Hicksville CNG Off Loading Facility	$90	$192	$102	Less work than planned
		Inwood CNG Off Loading Facility	$3,238	$3,746	$508	Less work than planned
		Northwest Nassau Transmission Main & Control Valve - Phase 2 & 3	$0	$2,250	$2,250	Work suspended due to COVID-19
		Pressure Regulating Facilities - SL-19	$0	$220	$220	Actuals charged to "Gas Reinforcement - LTLI12622 Hampton BaysSL-19 Rebuild". Budget from "Gas Reinforcement - LTLI12622 Hampton BaysSL-19 Rebuild" and "Pressure Regulating Facilities - SL-19" to be combined.
		Riverhead CNG Hardening Project	$3	$0	($3)	Unplanned work
		LTLI12021 Deer Park RCV 040632133	$3	$0	($3)	Minor variance not attributable to a specific cause
		LTLI12022 Pinelawn RCV 041025592	$3	$0	($3)	Minor variance not attributable to a specific cause
		LTLI2020 Deer Park RCV 040632166	$30	$0	($30)	Minor variance not attributable to a specific cause
Reliability			$39,956	$62,321	$22,364	
Non-Infrastructure	AMR	AMR Replacement	$1,558	$268	($1,290)	Budgeted under Automatic Meter Reading (AMR) - Growth and more units than planned
	AMR Replacement	Learning and Development - Materials, Tools and Equipment	$0	$108	$108	Fewer units than planned
	Purchase Misc Capital Equipment & Tools	Capital Equipment/Tools	$1,070	$1,371	$300	Fewer units than planned
		Meter Testing Equipment	$114	$89	($25)	More units than planned
		Telecomm - Comm site upgrades	$0	$34	$34	Minor variance not attributable to a specific cause
		Telecomm - Damaged Failure	$0	$9	$9	Minor variance not attributable to a specific cause
		Telecomm - Radio Capital Expenditures	$86	$35	($51)	More units than planned
	Special Projects	Special Projects (Operations Support)	($3)	$0	$3	Minor variance not attributable to a specific cause
Non-Infrastructure			$2,826	$1,913	($912)	

KeySpan Gas East Corporation
The Brooklyn Union Gas Company
d/b/a National Grid
Case 19-G-0309/0310
Attachment 1 to SANE-026
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KS GAS EAST CORP-KEDLI

CAPITAL BY CATEGORY - SEP CY2020 ($000's)

Classification	Category	Final Category Code	CY20 YTD - Actuals	CY20 YTD - Budget	CY20 YTD Variance to Budget	Spend Explanation
Other Actual	Other	Other	$0	($41,326)	($41,326)	
		Plug and Forecast Adjustments	$0	($2,719)	($2,719)	
Other Actual			**$0**	**($44,045)**	**($44,045)**	
Direct Capex (Incl. Cost of Removal)			**$278,214**	**$226,026**	**($52,189)**	
Total Cost of Removal		Cost Of Removal	**($19,150)**	**($7,809)**	**$11,342**	
Total Capital (Excl. Cost of Removal)			**$259,064**	**$218,217**	**($40,847)**	

BROOKLYN UNION GAS-KEDNY

CAPITAL BY CATEGORY - SEP FY2020 ($000's)

Classification	Category	Final Category Code	CY20 YTD - Actuals	CY20 YTD - Budget	CY20 YTD Variance to Budget	Spend Explanation
Growth	Base Growth	Automatic Meter Reading (AMR) - Growth	$0	$760	$760	Budgeted under "Automatic Meter Reading (AMR) - Growth." Actuals separately reported under AMR Replacement (Non-Infrastructure).
		Base Growth - Customer Contributions	($75)	($1,817)	($1,742)	Fewer credits than planned
		Base Growth - Install Meter / Regulator	$767	$754	($13)	Minor variance not attributable to a specific cause
		Base Growth - Meter Purchase / Operations	$0	$1,050	$1,050	Budgeted under "Base Growth - Meter Purchase / Operations." Actuals separately reported under Meter Purchases (Mandated).
		Build it Back Program	$105	($62)	($167)	Unplanned work
		Install Main	$13,124	$10,641	($2,483)	Higher unit cost
		Install Service	$29,251	$14,883	($14,368)	Higher unit cost
	Base Growth - Special Projects	GATEWAY DEVELOPMENT BROOKLYN	$880	($137)	($1,017)	Mischarges. Dollars will be moved to Install Services.
		LGA Delta Reconstruction	$51	($105)	($155)	Unplanned work
	Gas System Reinforcement	Gas Reinforcement - 16" ST on 39 St.	($66)	($108)	($42)	Timing of Actuals - work completed in prior period
		Gas Reinforcement - 16" ST on 77 St.	$221	($90)	($311)	Timing of Actuals - work completed in prior period
		Gas Reinforcement - 16" ST on East 9 St.	$0	($2)	($2)	Minor variance not attributable to a specific cause
		Gas Reinforcement - Chelsea Gate Station SI	$643	$180	($463)	Timing of Actuals - carryover of work from prior period
		Gas Reinforcement - Downtown Brooklyn Gate Station	$17	$96	$79	Timing of Actuals - work completed in prior period
		Gas Reinforcement - Gas Reinforcement - GRP 10789 -New	$0	$0	$0	
		Gas Reinforcement - Gov 116 to 60 psig inlet	$231	($154)	($385)	Timing of Actuals - carryover of work from prior period
		Gas Reinforcement - Gov 176 to 60 psig inlet	$0	$0	$0	
		Gas Reinforcement - GRP 10134-Gravesend Gate St rc	($4)	$0	$4	Minor variance not attributable to a specific cause
		Gas Reinforcement - GRP 10589-Lincoln Pl, Brooklyn	($0)	$0	$0	
		Gas Reinforcement - GRP 10778 - Dunlop Ave, Queens	($0)	$0	$0	
		Gas Reinforcement - GRP 10789 -New Gov, 236	$0	$0	$0	
		Gas Reinforcement - Kew Gardens Gate Station	$129	$1,102	$973	Timing of Actuals - Deferred work
		Gas Reinforcement - LTNY12284-Greenway N Ascan Ave-P	($0)	($2)	($2)	Minor variance not attributable to a specific cause
		Gas Reinforcement - LTNYXXXXX - Jamaica Inlet - PM	$0	$32	$32	Timing of Actuals - Deferred work
		Gas Reinforcement - New Gov in Crown Heights.	$0	$1	$1	Minor variance not attributable to a specific cause
		Gas Reinforcement - New Gov in Dyker Heights	($2)	($1)	$1	Minor variance not attributable to a specific cause
		Gas Reinforcement - New Gov in Flatbush	$0	$0	$0	
		Gas Reinforcement - New Gov in Ocean Pkwy	$0	$0	$0	
		Gas Reinforcement - New Gov in Prospect Leffert Gardens	$0	$26	$26	Minor variance not attributable to a specific cause
		Gas Reinforcement - New Gov in Sheepshead Bay	$0	$0	$0	
		Reinforcement	$10,530	$9,225	($1,305)	Timing of Actuals - Accelerated work
Growth			**$55,802**	**$36,272**	**($19,530)**	
			$0	$0	$0	
Mandated	City State Construction - Non-Reimbursable	CSC-Non-Reimbursables	$133,317	$52,365	($80,952)	More units than planned
	City State Construction - Reimbursable	CSC-Reimbursable	$65,180	$127,037	$61,857	Fewer units than planned
	City State Construction - Reimbursements	CSC-Reimbursements	($26,505)	($16,019)	$10,486	More credits than planned
	City State Construction - Special Projects	LaGuardia Redevelopment	$1,393	$488	($905)	More units than planned
		Land-LGA Gate Station Grand Central	$4	$0	($4)	Minor variance not attributable to a specific cause
		SE-851 -E.108 St Transmission Offse	$0	($4)	($4)	Minor variance not attributable to a specific cause
		SE851-Flatlands Ave Ph 1	$0	$0	$0	
		SE851-Flatlands Ave Ph 2	$9	($23)	($31)	Minor variance not attributable to a specific cause
		SE851-Flatlands Ave Ph 3	$0	$0	$0	
		SE851 - Flatlands Ph 4	$0	$0	$0	
		SE851- Flatlands Ph 5	$0	$0	$0	
		SE852- Flatlands Ph 4	$0	$0	$0	
		SE-853(Reimbursable)	$7,195	$1,613	($5,582)	Unplanned work
		SE-856 Transmission Offsets	$102	$0	($102)	Unplanned work
	Integrity	5.0.0.0.0.1 Launcher - Clove Lakes	$65	($17)	($82)	Unplanned work
		5.0.0.0.0.2;3;4 Receiver - Clove La	$4,719	($968)	($5,686)	Timing of Actuals - carryover of work from prior period
		Brooklyn BB S - 0.4;5;6 - Robotic I	$0	$0	($0)	
		Corrosion	$412	$511	$99	Work suspended due to COVID-19
		Cross Bore Remediation	$236	$205	($31)	Minor variance not attributable to a specific cause
		IMP-Southern Line Robotic ILI	$2,335	$2,500	$165	Timing of Actuals - work completed in prior period
		IVP Coney Island Spur Vlv1202 Re	$1,365	$1,351	($14)	Minor variance not attributable to a specific cause
		IVP Coney Island Spur Vlv 1216 Rep	$199	$115	($84)	Budgeted under "Pipeline Integrity Verification Coney Island Spur - Replace BQ 1216." Actuals separately reported under "IVP Coney Island Spur Vlv 1216 Rep" coupled with Minor Variance not attributed to a specific cause.
		IVP-Coney Isl Spur vlv-BQ1104 Repla	$33	$0	($33)	Minor variance not attributable to a specific cause
		Jamaica Bay - 0.2;3;4 - Robotic ILI	($482)	($2)	$479	Actuals reallocated to Jamaica Bay Line - 0.1 - Robotic IL from Jamaica Bay - 0.2;3;4 - Robotic ILI
		Jamaica Bay Line - 0.1 - Robotic IL	$1,343	$876	($467)	Actuals reallocated to Jamaica Bay Line - 0.1 - Robotic IL from Jamaica Bay - 0.2;3;4 - Robotic ILI & minor variance not attributed to a specific cause
		NY Joint Facilities	$0	($0)	($0)	
		Pipeline Integrity - IVP - Spur - 20" Spur to South Staten Isla	$0	$13	$13	Minor variance not attributable to a specific cause
		Pipeline Integrity (Transmission IMP)	$0	$60	$60	Budget amount assigned to individual Pipeline Integrity projects
		Pipeline Integrity Verification Coney Island Spur - Replace B	$0	$121	$121	Budgeted under "Pipeline Integrity Verification Coney Island Spur - Replace BQ 1216." Actuals separately reported under "IVP Coney Island Spur Vlv 1216 Rep."
		Restrictions for Elevated Gas Infrastructure	$0	$112	$112	Timing of Actuals - Deferred work
		Southern Line - East End Interconnect	$0	$637	$637	Timing of Actuals - Deferred work
		Spur to St. Albans GS ILI Enable	$0	$0	$0	
		Pipeline Integrity - IMP	$0	$250	$250	Work suspended due to COVID-19
		Pipeline Integrity - IVP	$0	$1,525	$1,525	Work suspended due to COVID-19
		Pipeline Integrity - IVP Reactive Main Replacement	$0	$250	$250	Work suspended due to COVID-19
		Transmission Station Integrity	$0	$1,125	$1,125	Timing of Actuals - Deferred work
		IMP-BB-Robot ILI Enable Newtown Cre	$10	$0	($10)	Minor variance not attributable to a specific cause
	Large Diameter CI Rehab	12th St - Lining Project	($3)	($34)	($31)	Minor variance not attributable to a specific cause
		69th St Liner - Lining Project	$0	$756	$756	Timing of Actuals - Deferred work
		Cast Iron Main Lining-NYC	$1,854	($32)	($1,886)	Mischarges. Dollars will be moved to CI Lining - 57th Ave, Queens
		CISBOT	$1,661	$4,120	$2,459	Fewer units than planned
		Lining Prj - Pacific St/Junius St	($0)	$866	$866	Timing of Actuals - work completed in prior period
		Lining Proj. 61 St	$240	$1,302	$1,062	Timing of Actuals - Deferred work
		Lining Proj. Belmont Ave	$628	$3,018	$2,391	Timing of Actuals - Deferred work
		Lining Project - 101st Ave	($0)	$117	$117	Timing of Actuals - work completed in prior period
		Lining Project - 150th St, Queens	$10	$498	$488	Timing of Actuals - work completed in prior period
		Lining Project - 4th Ave, Brooklyn	$0	($0)	($0)	
		Lining Project - 65th St	($0)	$48	$49	Minor variance not attributable to a specific cause
		Lining Project - 69th St, Queens	$1,371	$4,097	$2,726	Timing of Actuals - Deferred work
		Lining Project - 76th St	$1	($0)	($1)	Timing of Actuals - work completed in prior period
		Lining Project - Colfax St	$0	$185	$185	Timing of Actuals - work completed in prior period
		Lining Project - Kent Ave	($0)	($6)	($6)	Minor variance not attributable to a specific cause
		Lining Project - Springfield Blvd	$0	($1)	($1)	Minor variance not attributable to a specific cause
		CI Lining - 134th St, Queens	$2,424	$55	($2,369)	Timing of Actuals - Accelerated work
		CI Lining - 57th Ave, Queens	$569	$1,550	$981	Actuals mischarged to Cast Iron Main Lining coupled with Timing of Actuals - Accelerated work
		CI Lining - Ave Y, Brooklyn	$820	$55	($765)	Timing of Actuals - Accelerated work
	Local Law 30	LL30	$16,843	$12,485	($4,358)	Accelerated work and Higher unit cost
	Main Replacements - Proactive Programs	12th St - Main Replacement	$0	($232)	($232)	Unplanned work
		Avenue U, Brooklyn	$0	($1)	($1)	Minor variance not attributable to a specific cause
		Main Repl Leak Prone Pipe	$82,331	$129,217	$46,886	Work suspended due to COVID-19 & Timing of Actuals - work completed in prior period
	Main Replacements - Reactive Programs	Maintenance	$1,729	$2,725	$997	Fewer units than planned
	Meter Work	Meter Changes	$1,609	$3,266	$1,657	Fewer units than planned
		Other Meter Work	$120	($22)	($143)	Non-reoccurring reactive work related to Meter Change program. Unplanned work related to Meter Changes that was recorded separately.
	Purchase Meters (Replacements)	Meter Purchases	$4,276	$2,525	($1,751)	Partial budgeted in "Base Growth - Meter Purchase / Operations." Actuals separately reported under Meter Purchases (Mandated) & More units than planned.
	Service Replacements - Proactive Programs	Svc Repl-BS HP Leak Prone Services	$337	$1,308	$970	Fewer units than planned
	Service Replacements - Reactive Programs	Atmospheric Corrosion/Inside Inspections	$783	$3,206	$2,423	Work suspended due to COVID-19
		Latent Damages	$0	$0	$0	
		Latent Damage Verification	$0	$136	$136	Fewer units than planned
		Leaks	$1,887	$4,491	$2,603	Fewer units than planned
		Non-Leak / Other	$3,390	$3,210	($180)	More units than planned
		Soft-Off Process (Cutoffs)	$137	$148	$10	Minor variance not attributable to a specific cause
		Valve Installation/Replacement	$0	$85	$85	Work suspended due to COVID-19
	Special Projects	Gas Infrastructure Replacement on Structures	$226	$0	($226)	Timing of Actuals - carryover of work from prior period
		High Density Polyethylene Services	$0	$922	$922	Funds reallocated to Install Main, LPP, CSC, Reinforcement, Maintenance, Water Intrusion, and Gas Reliability
		Low Pressure Main Valve Installation	$0	$923	$923	Funds reallocated to Install Main, LPP, CSC, Reinforcement, Maintenance, Water Intrusion, and Gas Reliability
		Operator Qualification Program	$0	$455	$455	Work suspended due to COVID-19
		Plastic Fusion - In Process Inspections	$0	$115	$115	Funds reallocated to Install Main, LPP, CSC, Reinforcement, Maintenance, Water Intrusion, and Gas Reliability
		Plastic Fusion QA/QC Re-Digs	$0	$1,219	$1,219	Funds reallocated to Install Main, LPP, CSC, Reinforcement, Maintenance, Water Intrusion, and Gas Reliability
Mandated			**$314,174**	**$356,898**	**$42,724**	

BROOKLYN UNION GAS-KEDNY

CAPITAL BY CATEGORY - SEP FY2020 ($000's)

Classification	Category	Final Category Code	CY20 YTD - Actuals	CY20 YTD - Budget	CY20 YTD Variance to Budget	Spend Explanation
Reliability	CNG Programs	CNG - NY Brooklyn (Canarsie) - Compressor Upgrade, New	$0	$42	$42	Minor variance not attributable to a specific cause
		CNG - KEDNY Blanket	$0	$124	$124	Work suspended due to COVID-19
		CNG - KEDNY Contract Closeout	$0	$200	$200	Work suspended due to COVID-19
	CNG Programs - Special Projects	STATEN ISLAND CNG FILLING STATION	$108	$15	($93)	Higher unit cost
	Gas System Reliability	CNG - NY Brooklyn (Greenpoint) Fuel	($5)	$30	$34	Minor variance not attributable to a specific cause
		CNG - NY Brooklyn (Greenpoint) New	$14	$485	$471	Timing of Actuals - Deferred work
		Gas Planning/RCV Program	$2,934	$1,406	($1,528)	Higher unit cost
		Gas System Reliability - Gas Control (Training Simulator)	$0	$28	$28	Timing of Actuals - Deferred work
		GOV-107 Refurbish	$162	$536	$373	Timing of Actuals - Deferred work
		GOV 228 Retirement	$0	$0	$0	
		GOV 305 Refurbish Station	$177	$58	($119)	Timing of Actuals - Accelerated work
		Gov 54 Refurbishment	$11	$0	($11)	Minor variance not attributable to a specific cause
		Gov 68 Refurbishment	$13	$0	($13)	Minor variance not attributable to a specific cause
		GRS Heater Program-NYC	$56	$164	$108	Work suspended due to COVID-19
		LTNY10240 - Grasmere Reliability	$0	$48	$48	Minor variance not attributable to a specific cause
		PRE-Fresh Kills Methane Recovery	$0	$0	$0	
		PRE-IF-04/ IF-07	$1,869	$709	($1,160)	Timing of Actuals - carryover of work from prior period
		PRE-PR GOV211 86th/Ft HAMPWY	$2	$0	($2)	Minor variance not attributable to a specific cause
		PRE-Shafer Vlv Narrows Upgrade	$433	$286	($146)	Timing of Actuals - Accelerated work
		Pres Reg Facil - proactive-NYC	$20	$2,183	$2,163	Fewer units than planned
		Pres Reg Facil - Reactive-NYC	$361	$406	$45	Fewer units than planned
		Pressure Reg CNG-Proactive-NYC	$35	$113	$78	Fewer units than planned
		Pressure Reg CNG-Reactive-NYC	$0	$166	$166	Fewer units than planned
		Pressure Regulating Facilities - GOV - 54	$0	$33	$33	Timing of Actuals - Deferred work
		Pressure Regulating facilities - Gov-68	$0	$33	$33	Timing of Actuals - Deferred work
		Storm Hardening Install remote Service Shutoff Valves	$184	$1,096	$912	Timing of Actuals - Deferred work
		System Automation	$652	$802	$149	Timing of Actuals - Deferred work
		Gov 303	$54	$55	$1	Minor variance not attributable to a specific cause
		Distribution Station Over Pressure Protection	$0	$348	$348	Timing of Actuals - Deferred work
		Pressure Regulating Facilities - GOV 303	$0	$30	$30	Minor variance not attributable to a specific cause
		Tetco Heater #3 Replacement	$38	$45	$7	Minor variance not attributable to a specific cause
		Water Intrusion	$9	$261	$252	Fewer units than planned
	LNG Programs	LNG - Boiloff Heaters/Steam Boiler Upgrade	$0	$253	$253	Less work than planned
		LNG - Controls System Upgrade	$0	$199	$199	Less work than planned
		LNG - Generator Stepup Transformer	$0	$3	$3	Minor variance not attributable to a specific cause
		LNG - Greenpoint LNG	$1,769	$2,320	$551	Fewer units than planned
		LNG - Piping Insulation Upgrade-NY	$120	$0	($120)	Timing of Actuals - Accelerated work
		LNG - Hi Ex Foam System	$0	$17	$17	Minor variance not attributable to a specific cause
		LNG - Liquefaction Critical Spares	$17	$483	$466	Work suspended due to COVID-19
		LNG - Pipe Supports & Insulation	$0	$17	$17	Minor variance not attributable to a specific cause
		LNG - RNG Blanket	$0	$76	$76	Timing of Actuals - Deferred work
		LNG - Replacement of Vaporizer 7 & 8	$0	$33	$33	Timing of Actuals - Deferred work
	LNG Programs - Special Projects	Geoweb Dike Replacement	$488	$56	($432)	Timing of Actuals - carryover of work from prior period
		Hi Ex Foam System	$10	$143	$133	Timing of Actuals - Deferred work
		Hydrant & Deluge Piping Upgrade	$359	$635	$276	Timing of Actuals - Deferred work
		Ice Shield	$0	$0	$0	
		LNG - Vaporizers 13 and 14	$8,688	$11,901	$3,213	Timing of Actuals - Deferred work
		Relocate Maintenance Area	$312	$827	$515	Less work than planned
		Salt Water Pump House Upgrade	$2,593	$2,678	$85	Timing of Actuals - Deferred work
		Tail Gas Compressor Upgrade	$102	$450	$347	Timing of Actuals - Deferred work
		Truck Load/Unload Station	$2,126	$1,475	($650)	Higher unit cost
		Clove Lakes Transmission Line Loop	$0	$250	$250	Less work than planned
		LNG - Bulkhead Upgrade	$0	$350	$350	Timing of Actuals - Deferred work
		LNG - Plant Outlet Drip Leg	$0	$5	$5	Minor variance not attributable to a specific cause
		Vaporizer 3 & 4 Replacement	$5,748	$5,905	$157	Timing of Actuals - Deferred work
		Controls System Upgrade	$3	$192	$190	Less work than planned
		Vaporizer 7 & 8 Replacement	$0	$0	($0)	
	Special Projects	Bay Ridge Gate Station Refurbishmnt	($23)	($98)	($75)	Timing of Actuals - work completed in prior period
		Bowery Bay Station Upgrade	$0	$33	$33	Timing of Actuals - Deferred work
		BQI - Phase III	$0	($0)	($0)	
		Buried Vent Line Remediation NY	$0	$0	$0	
		Buried Vent Line Remediation NY	$95	$42	($54)	Unplanned work
		CGT-Citizen Gate Tunnel	$3,460	$2,484	($976)	More work than planned
		Citizens Gate Bulkhead Replacement	$3,608	$6,382	$2,774	Lower unit cost
		Floyd Bennett Field M&R ROV's	$0	($2)	($2)	Minor variance not attributable to a specific cause
		Gas Control Telemetry Upgrade 3G/4G	$4	$61	$56	Timing of Actuals - work completed in prior period
		Gas System Control - DNY	$0	$82	$82	Timing of Actuals - Deferred work
		KEDNY CNG Facility Site 1	$4,167	$25	($4,142)	Mischarge
		KEDNY Sandy Recovery Program	$0	($77)	($77)	Budget adjustment
		LTNY11044 - RCV BQ1243 - PM	$0	$33	$33	Minor variance not attributable to a specific cause
		LTNY13231-Marine Park Regulator-PM	$150	$25	($124)	Timing of Actuals - carryover of work from prior period
		M2M Upgrade	$0	$174	$174	Timing of Actuals - Deferred work
		MRI - Metro Reliability Inf	$0	$13	$13	Minor variance not attributable to a specific cause
		MRI-Metro Reliability Inf	$57,063	$54,106	($2,957)	Timing of Actuals - Accelerated work
		Newtown Creek Renewable Gas Project	$5,175	$3,573	($1,602)	Timing of Actuals - Accelerated work
		North Brooklyn Mini Gate	$149	$30	($119)	Timing of Actuals - Accelerated work
		Northern Queens T&D reinf.	($258)	$134	$393	Budget adjustment to OPEX.
		PRE-Coney Island Heater + Mini gate	$0	($10)	($10)	Minor variance not attributable to a specific cause
		PRE-SP-Maspeth St Decommissioning..	$0	$0	$0	
		Spring Creek Reconfiguration	$1,085	$1,411	$326	Lower unit cost
		Tetco/Spectra Goethals Relief Vlv	$1,423	$266	($1,157)	Higher unit cost
		Greenpoint CNG Off Loading Facility	$749	$425	($324)	Timing of Actuals - Accelerated work
		Greenpoint Trans to LNG	$4	$0	($4)	Minor variance not attributable to a specific cause
		Hyman Station Refurbishment	$39	$45	$6	Minor variance not attributable to a specific cause
		LTNY11690 - LGA Backfeed - PM	$0	$28	$28	Minor variance not attributable to a specific cause
		Renewable Natural Gas (RNG) Interconnections	$0	$450	$450	Fewer units than planned
		Varick Compressor Bld Retirement	$160	$328	$168	Lower unit cost
Reliability			**$106,512**	**$107,933**	**$1,421**	
Non-Infrastructure	AMR	AMR Replacement	$3,435	$2,121	($1,314)	Partial budgeted in "Automatic Meter Reading (AMR) - Growth." Actuals separately reported under AMR Replacement (Non-Infrastructure) and Timing of Actuals - Accelerated work.
	AMR Replacement	Learning and Development - Materials, Tools and Equipment	$0	$70	$70	Budget adjustment to Gas Training Capex Tool Budg-KEDNY
	AMR - Special Projects	AMR Installation	$1,725	$1,505	($220)	More units than planned
	Purchase Misc Capital Equipment & Tools	Capital Equipment/Tools	$2,533	$2,592	$59	Fewer units than planned
		Gas Training Capex Tool Budg-KEDNY	$22	$70	$48	Partial budgeted in Learning and Development - Materials, Tools and Equipment coupled with Work suspended due to COVID-19
		Meter Testing Equipment	($74)	$90	$164	Budget adjustment
		Telecomm - Damaged Failure	$0	$4	$4	Minor variance not attributable to a specific cause
		Telecomm - Radio Capital Expenditures	$46	$50	$3	Minor variance not attributable to a specific cause
	Special Projects	Special Projects (Operations Support)	$0	$0	$0	
Non-Infrastructure			**$7,688**	**$6,503**	**($1,185)**	
Other Actual	Other	Other Actuals	$0	($54,298)	($54,298)	Budget Adjustment
		Other Forecast Adjustments	$0	$0	$0	
Other Actual			**$0**	**($54,298)**	**($54,298)**	
Direct Capex (Incl. Cost of Removal)			**$484,176**	**$453,308**	**($30,868)**	
Total Cost of Removal		**Cost Of Removal**	**($38,212)**	**($21,149)**	**$17,063**	
Total Capital (Excl. Cost of Removal)			**$445,964**	**$432,159**	**($13,805)**	

LEAK PRONE PIPE (miles) - KEDNY

	<<< 2020 Calendar Year >>>												Year to Date
	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	CY'20
K E D N Y L P P													
Main Replacement Program	3.3	1.6	1.5	0.3			0.1	0.1	2.1				8.9
City/State Construction	2.4	2.5	2.2	0.3	0.2	1.1	2.9	2.0	5.6				19.0
Leak / Reactive misc. Mandated	0.0	-	0.0	-	-	-	-	-	-				0.0
Growth Reinforcement	0.5	-	-	-	-	-	-	-	0.4				0.8
Reliability/Water Intusion	-	-	0.0	-	-	-	-	-	-				0.0
Actual Total (FCOMP, COMP or CASBUILT	6.2	4.0	3.7	0.6	0.2	1.1	3.0	2.0	8.0	-	-	-	28.8

LEAK PRONE PIPE (miles) - KEDLI

	<<< 2020 Calendar Year >>>												Year to Date
	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec	CY'20
K E D L I L P P													
Main Replacement Program	14.2	5.0	9.9	0.8	0.2	0.1	8.9	6.4	14.5				60.0
City/State Construction	0.1	-	0.6	0.2	-	-	-	-	-				0.9
Leak / Reactive misc. Mandated	-	0.0	0.1	-	-	0.0	-	-	0.2				0.3
Growth Reinforcement	-	-	-	-	-	0.0	-	-	1.3				1.3
Reliability/Water Intusion	-	-	-	-	-	-	-	-	-				-
Actual Total (FCOMP, COMP or CASBUILT)	14.33	4.99	10.59	0.95	0.16	0.17	8.89	6.42	15.97	-	-	-	62.5